STANDARDS
&
SPECIFICATIONS

FOR THE

DISN Enterprise Network (DISN-EN)

INSTALLATION CAMPUS AREA NETWORK DESIGN & IMPLEMENTATION (ICAN-DI)

FOR THE

IMPLEMENTERS OF THE DISN-EN INFRASTRUCTURE

Artifact Version 1.09f FINAL
(5 December 2016)

The DISN-EN *(Enterprise Network)* ICAN-DI *(Design and Implementation)* Standards & Specifications facilitate the systematic, stable and sustainable IT Infrastructure instantiation of Base/Post/Camp/Stations that enables the Joint Enterprise Vision. This artifact lays out the ICAN technology portion/segment of the overall DISN-EN Architecture for implementers. This artifact supersedes all previously issued ICAN implementation standards and specifications *(i.e.: TR No. ELIE-ICE-TI 10-001 and DISN-EN ICAN-DI Standards and Specifications v1.08)*. Adoption of these standards and specifications is done on a service agency-by-service agency basis in support of a unified whole, and is currently adopted as the ICAN Implementation Standards and Specifications by:

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DOCUMENT REVISION TRACKING

Disclaimer: The user of this document is responsible for obtaining the latest document revision released for use on the DISN-EN. Documents released for use are maintained under configuration management control and are archived in the configuration management library. Associated document revisions will be made in accordance with the appropriate Configuration and Quality Management processes. Configuration management control will be maintained until the document is downloaded from the configuration management library and either stored locally to a personal computer (PC), printed, or copied. Documents that are no longer under configuration management control are to be treated as For Official Use Only (FOUO) in accordance with DISN-EN security procedures. Documents superseded by later revisions are designated obsolete and are no longer appropriate for use on the DISN-EN.

CR #	REVISION	Summary of Change
	Version 1.00	2/11/2014: First Non-Draft Release
	Version 1.01	4/18/2014: RFI Response, Market Research, and Collaborations with I3MP and P2E to modify Appendix-A
		Remove Next-Day Replacement Requirement (moves to SOW)
		Remove ESA Exemption (moves to SOW)
		Modify Phrase “Industry Standard CLI” to simply “CLI”
		Modify ACS Max Port Count 10G Line Rate (330 to 256)
		Exchange the phrase “oversubscription” to “non-blocking”
		Remove the MACsec requirements from ACS and ADS Devices
	Version 1.02	4/23/2014: Market Research; Updates to Appendix A
		Modify 1Gbps/10Gbps Dual-Speed Requirement (Only 10Gbps Required)
		P2E Recommended Nomenclature Changes (Cosmetic)
		Update terminology of “E-PE” to “JB-CE” (Sync up with MPLS Standards)
	Version 1.03	5/5/2014: Revised Signature Page (From Mr. Bezwada to Mr. Blohm)
	Version 1.04	6/25/2014: NETMOD-C Surveys
		Add Fiber SFP-based Switches to EAS Section of Appendix A
		Separate Appendix A from the Main ICAN-DI to allow easy reference
	Version 1.05	7/28/2014: Draft RFQ Responses
		Limit 802.1AE MACsec requirement for downlink EAS ports only
		Change ADS certification to ASLAN-Certified
	Version 1.06	8/18/2014: RFQ Response
		Remove MACsec requirement from L2-BS switch specifications
	Version 1.07	8/20/2014: Rephrases section 1.5.2.1 (applicable to min required ports only)
	Version 1.08	8/28/2014: Modified section 1.7.2 (Mgmt. Port Requirement set to 10/100 Mbps)

CR #	REVISION	Summary of Change
	Version 1.09	6/10/2016: Updated Artifact with PEO EIS CIO ERB Change Requests
		Version Revision & Outcome from PEO EIS CIO ERB/CCB Meeting
		In light of Tier-1 vendors now being able to support hostnames longer than 32 characters, Hostname Max Length and Structure has been made less Restrictive.
		Legitimized pervasively implemented but not previously approved use of native vendor First Hop Gateway Redundancy protocols instead of strict use of VRRP.
		Legitimized pervasively implemented but not previously approved use of "L2BS" instead of "b01" in hostnames of L2 Bridging Switches.
		Legitimized pervasively implemented but not previously approved use of *".agency.mil"* suffix to the names of the MST instances, designating the executive agent of the B/P/C/S.
		Legitimized pervasively implemented but not previously approved out of order use of port descriptions. The order has been changed to fit what is in the field.
		Clarified hostname suffix, (-1 vs. -1s1)
		Changed Multicast Ingress Port Policy Limit from 20% to 5%
		Removed Q-in-Q as a solution to L2 tunneling (scalability problem identified)
		Updated Sections & Diagrams – Reflective of Most Recent Naming/Concepts
		Updated ICAN Switch Technical Requirements (added SDN Compatibility)
		Added Numbering to Sections, Diagrams and Appendix for Easier Reference
		Updated Glossary and Acronym List
	Version 1.09c	9/7/2016: Updated Artifact with NETCOM Provided Change Requests:
		Added language to reinforce the Joint nature of the document.
		Expanded Do's/Don'ts section to further clarify intended architecture.
		Updated VLAN Numbering Schema to reflect more granular details.
		Updated Port Configuration Standards to capture most recent STIG requirements.
		Refined Layer 2 Extension language to enhance clarity.
		Enhanced guidance on labeling standards.
		Updated Appendix to reflect main document changes.
	Version 1.09d	9/20/2016: Removed MACsec requirement for EAS downlink ports in Appendix.
	Version 1.09e	10/20/2016: Updated per vendor questions. VLAN limit of 4094, Minimum Jumbo Frame size of 9000bytes
		Added Small Form Factor EAS requirements in 7.8
	Version 1.09f	11/22/2016: Updated hostname guidance to reflect decision by NETCOM to precede the hostname with the device make and model.
		Added HAIPE/CT to VLAN chart per request.

Revision Policy: This document is revised and updated as needed through the collective agreement of the Service Agencies affected by these standards.

1.0 SCOPE

The scope of this document is focused on providing relevant guidelines, standards and specifications as they pertain to the Defense Information Systems Network - Enterprise Network (DISN-EN) Architecture. This artifact is intended to service Army, Air Force, Marines and the Navy by providing a scalable means of standardizing Base/Post/Camp/Station infrastructure and infrastructure configurations. This artifact is a key component in collectively reaching Enterprise maturity under a unified management framework and thereby the pathway toward achieving the Joint Information Environment (JIE). This artifact is intended to be a living document with need for periodic updates, i.e.: when lifecycle induced technological leaps are introduced into the architecture.

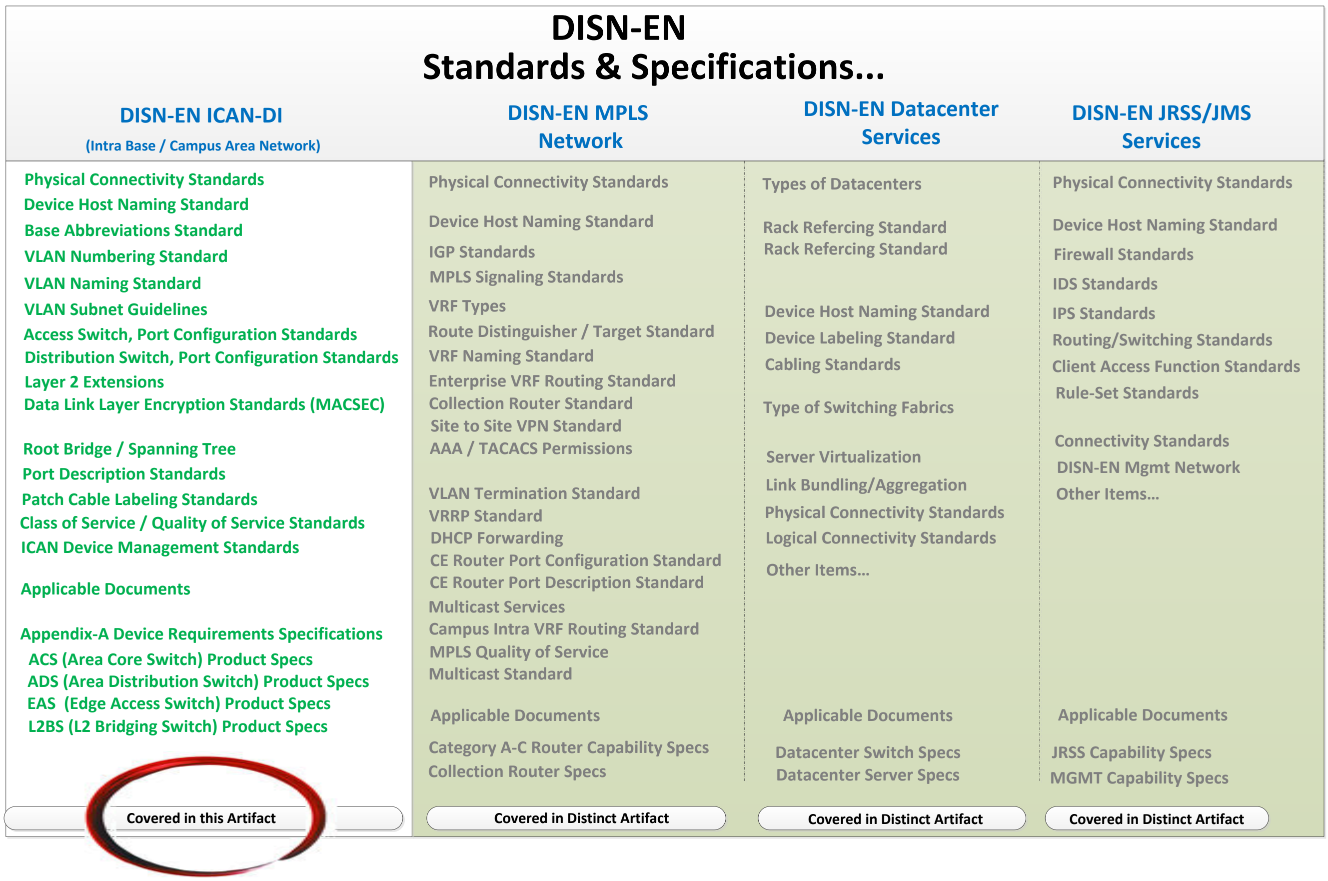


Diagram 1.0 – ICAN-DI Contents

2.0 INTRODUCTION

The DISN-EN is a multi-agency solution that creates an Enterprise Information Technology (IT) Environment across all participating agencies relying on a standardized core Internet Protocol (IP) transport service network (Multi-Protocol Label Switching (*MPLS) Service Network*), with regionally collapsed security stack services (*Joint Regional Security Stacks (JRSS)*) and a standardized Base/Post/Camp/Station (B/P/C/S) campus area distribution network (Installation Campus Area Network (ICAN)). The DISN-EN capability infrastructure advocates for a Joint Management solution (*DISN-EN Management*) that delivers true multi-tenancy in a shared operating environment. All of these components are intended to work together in a dovetailed manner in order to realize the Enterprise vision (active steps toward the Joint Information Environment (JIE)).

2.1 Installation Campus Area Networks

The ICANs provide high speed local IP switched transport services within the B/P/C/S, from the Top Tier Base Nodes (TTBNs - first hop routers) down to the user access devices like desktops and printers. The ICAN infrastructure follows a classic framework of Core, Distribution and Access and is therefore comprised of Area Core Switches (ACS), Area Distribution Switches (ADS) and Edge Access Switches (EAS). The DISN-EN architecture observes a highly stable systematic hierarchy for physical and logical interconnectivity between ACS', ADS' and EAS'. Each ACS Pair creates a physically distinct Layer 2 Domain that services the layer 2 production VLANs within that network segment or region. B/P/C/S' may have as many ACS Pairs as they reasonably need to meet the density demands of the ICAN. In instances where a single VLAN must be shared across multiple L2 Domains for whatever reason, a tightly controlled Layer 2 Bridging Switch (L2BS) device is to be used to ensure that the network remains loop free and stable. The standards and technical specifications as it relates to the "ICAN" are captured in this artifact.

2.2 DISN-EN MPLS Service Network

The DISN-EN MPLS Service Network provides unclassified Wide Area Network (WAN) label switched transport connectivity within and between service regions across the globe. IP Transport within and between service regions utilize a layered MPLS solution in order to preserve the Defense Information Systems Agency (DISA) circuit provider transport backbone (Inner MPLS Core) and keep it logically and administratively distinct from the Common Services Transport Backbone (Outer MPLS Core). The Common Services Transport Backbone is where the ICAN traffic, Datacenter traffic, the JRSS and Joint Management System (JMS) services reside... building toward the Joint "Common" Information Environment. The "Inner Core" is primarily comprised of DISA Provider nodes (P-nodes) and DISA IP Transport Provider Edge (IPT-PE) routers interconnected through an extensive fiber optic/Dense Wavelength Division Multiplexing (DWDM) infrastructure. The "Outer Core" is primarily comprised of the Joint Regional Customer Edge (JR-CE) routers, the Joint Base Customer Edge (JB-CE) routers and the Joint Base Collection Router (JB-CR) routers. The standards and technical specifications as it relates to the "Outer Core" are captured in a distinct artifact.

2.3 DISN-EN Joint Regional Security Stacks

The JRSS' provide network traffic intrusion detection, intrusion prevention, multi-tier firewall services, content filtering, logging, packet capture services and inter Virtual Routing and Forwarding (VRF) policy filtered routing services for all participating B/P/C/S. The JRSS has been designed to provide these security services to all participating Department of Defense (DoD) Agencies no matter what B/P/C/S they reside on within a JRSS' regional boundary. The JRSS has been accepted by the DoD - Chief Information Officer (CIO) and DISA as a significant part of the instantiation of the Single Security Architecture envisioned to realize the JIE. Inter-regional IP traffic traverses JRSS control points enabling United States Cyber Command (USCYBERCOM) end-to-end cybersecurity visibility and control. In relation to the ICAN, the JRSS is where logically separated security zones are permitted to change security contexts (be routed together), assuming the individual security zone policies permit such activity. The intent of the JRSS' is to enable enterprise security services for DoD and de-duplicate locally operated security stacks across DoD (previously identified as Top Level Aggregation (TLA) stacks). The standards and technical specifications as it relates to the "JRSS" are captured in a distinct artifact.

2.4 DISN-EN Joint Management Capability

The JMS provides DISN-EN Fault, Configuration, Accounting, Performance, Security and Capacity Management. It is the first instantiation of the envisioned JIE Enterprise end-to-end management solution that relies on shared management platforms through virtualization and enterprise level role based access (moving away from duplicative local management systems). JMS reaches its managed elements via the Joint Management Network

(JMN) which is a hybrid in-band/out-of-band transport network augmenting the DISN-EN, ensuring critical items are always accessible to its operators. Managed elements that are configured with in-band only management connectivity (i.e.: an access switch), accept the risk of rare but possible temporary loss of management connectivity by design due to a cost benefit analysis that indicates that reserving or pulling separate fiber to such a device is cost prohibitive. JMS provides a portal based interface to enable management views from anywhere to authorized operators and provides delegated access to trusted operators on devices they are functionally authorized to operate. The standards and technical specifications as it relates to the "JMS" and "JMN" are captured in a distinct artifact.

3.0 INSTALLATION CAMPUS AREA NETWORKS

3.1 Introduction

The Installation Campus Area Networks (ICANs) provide the campus distribution IP connectivity from the TTBNs down to the user access devices like desktops and printers. The ICAN infrastructure is comprised of Area Core Switches (ACS), Area Distribution Switches (ADS), where they are deemed appropriate by the Executive Agent of the base, and Edge Access Switches (EAS) directly servicing users, printers and other networked devices. The DISN-EN architecture observes a systematic hierarchy for interconnectivity between ACS', ADS' and EAS'. Each ACS Pair will create a physically separate Layer 2 Domain, where inter-area layer 2 VLAN extensions would be handled through a tightly controlled L2BS device.

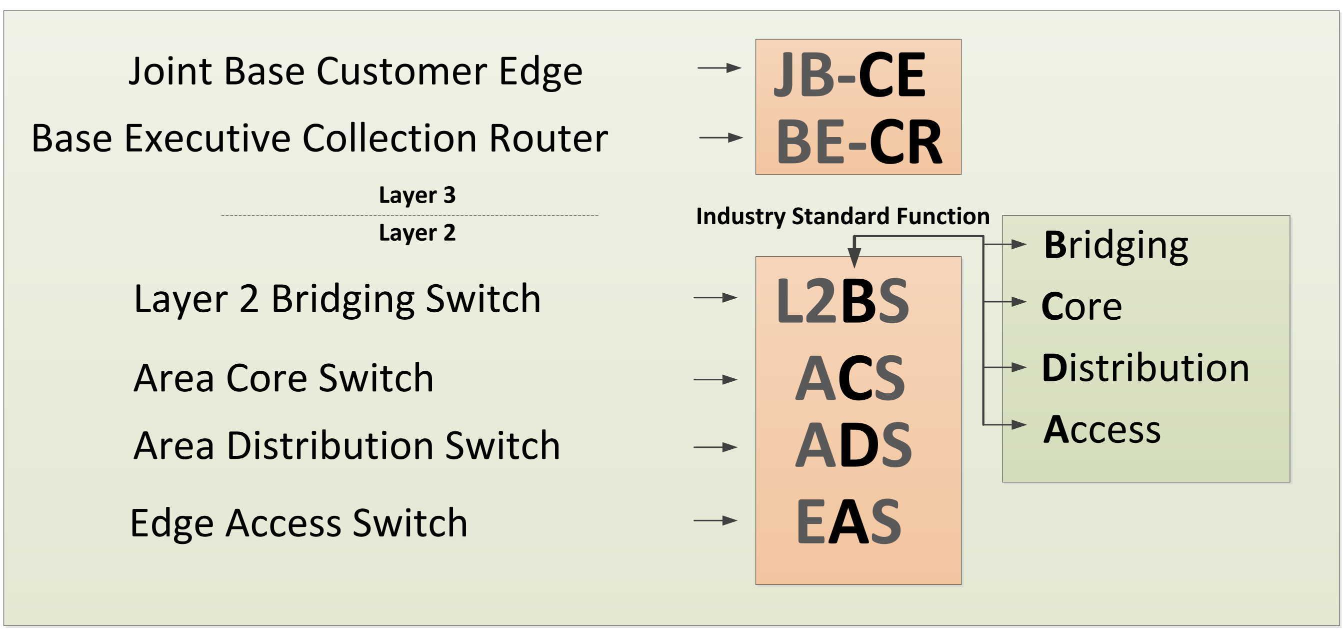


Diagram 3.1 – ICAN Acronyms

3.2 Description of Intent

The intent of the ICAN is to establish a clear understanding of the planned campus area delivery network infrastructure on B/P/C/S through the use of L2BS, ACS, ADS and EAS devices, to include any remote instantiations of ACS' and EAS' logically attached to a local ICAN, framed within defined device requirements and configuration standards. A remote ACS follows the same hierarchical physical and logical architecture as a typical ACS, however it is physically outside the boundaries of the normal B/P/C/S infrastructure yet it is physically connected back to the JB-CE routers at the parent B/P/C/S for Layer 3 services. A remote ACS offers infrastructure flexibility to implementers so that they don't have to have a set of Layer 3 aggregation devices at a remote (small/medium) size site. External ACS locations are locations that service a single L2 Domain, but do not have a JB-CE pair of routers at that location. A good example of this would be sites that are NOT already on the DISA backbone and connect to a router on another installation for access to the DISN-EN. A remote EAS is also provided as an option for mini remote sites, comprised of a single switch or stack of switches. Remote EAS devices uplink to the Base Executive Collection System for their first hop routing. Please note that all interconnectivity between a remote and its parent site, must adhere to relevant transport security (encryption) standards.

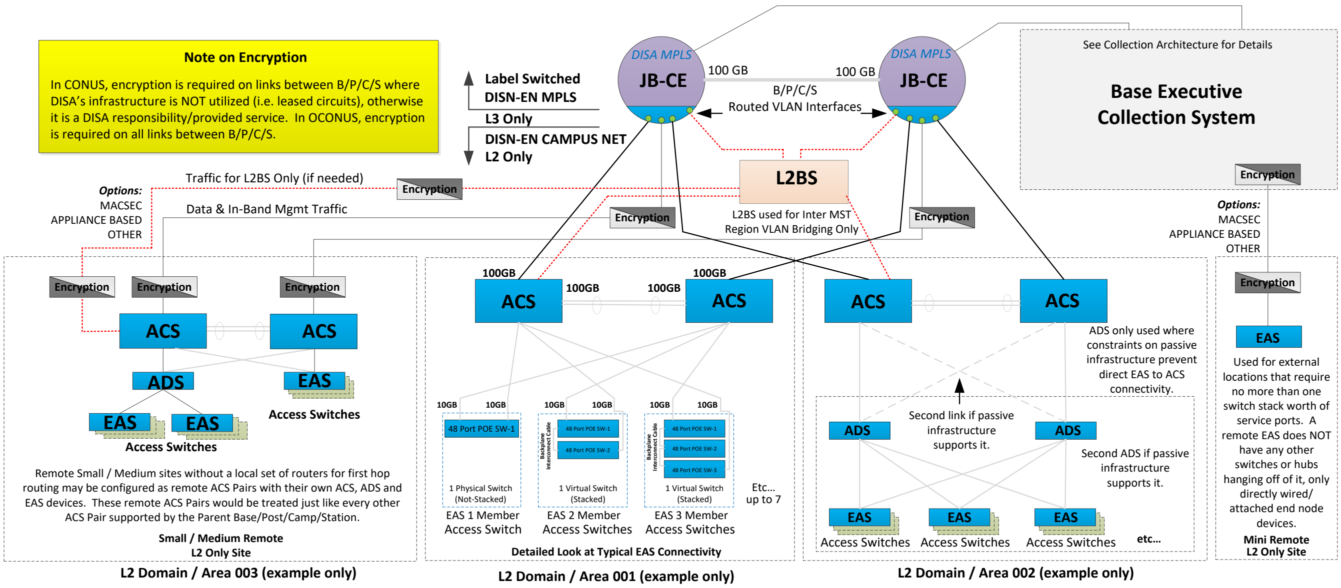


Diagram 3.2 – Overall Installation Campus Area Network (ICAN) Diagram

3.3 Physical Connectivity Standards

The ICAN is comprised of not only the active (switches, routers and encryption) components but also the existing physical cable plant available to the local B/P/C/S. It is recognized that the ICAN inherits its existing B/P/C/S' current level of physical cable plant, space and power limitations. Due to this known set of limitations, the ICAN administrators will require flexibility in how their networks are to be implemented.

The table below provides the maximum number of recommended EAS switches in a given switch stack. Both an availability impact calculation as well as the target bandwidth availability calculation per user point to the same recommended maximum number, mainly that there ought to be no more than seven (7) 48 port switches stacked in any given stack. While most vendors allow more switches to be added to the stack from a technical perspective, it undermines availability impact risk management and per user target bandwidth availability when the recommended maximum number is exceeded.

Availability Impact Analysis		
Uplink Bandwidth of Stack:	10	Gigabits
Ports per Switch in Stack:	48	Ports
Number of Switches in Stack:	7	Switches
Target Availability:	99.99%	UCR
Avg User Ports on Base:	22,633	People
Total Avg User Hours/Yr:	198,265,080	Hours
Per Person Allowed Outage/Yr:	52.56	Minutes
Total Outage Time Permited/Yr:	19,827	Hours
Estimated Avg Outage Time:	30	Minutes (UCR)
Possible Number of Users/Stack:	336	Possible Users
Outage Impact:	168	Hours
Per Incident Impact to Outage Time Permitted/Yr:	0.84735%	
Percent of EAS' that may experience Outage/Yr:	0.34	Target is no less than 1/3 of stacks

Bandwidth Impact Analysis		
Uplink Bandwidth of Stack:	10	Gigabits
Ports per Switch in Stack:	48	Ports
Number of Switches in Stack:	7	Switches
Target Bandwidth/User:	30	Mbps
HD Video Stream:	8.38	Mbps
Phone Traffic:	0.10	Mbps
User Data Transfere Traffic:	10.00	Mbps
Total Bandwidth Needed /User:	18.48	Mbps
Possible Number of Users/Stack:	336	Possible Users
Total Uplink Bandwidth Target:	9.84	Gigabits
Total Uplink Bandwidth Needed/Stack:	6.07	Gigabits
EAS Bandwidth Utilization (Daily Use):	6.56250%	
EAS Bandwidth Utilization (Max Use):	60.65063%	

Diagram 3.3.1 – Stacking Limit, Availability & Bandwidth Analysis

The Do and Don'ts of Physical Connectivity

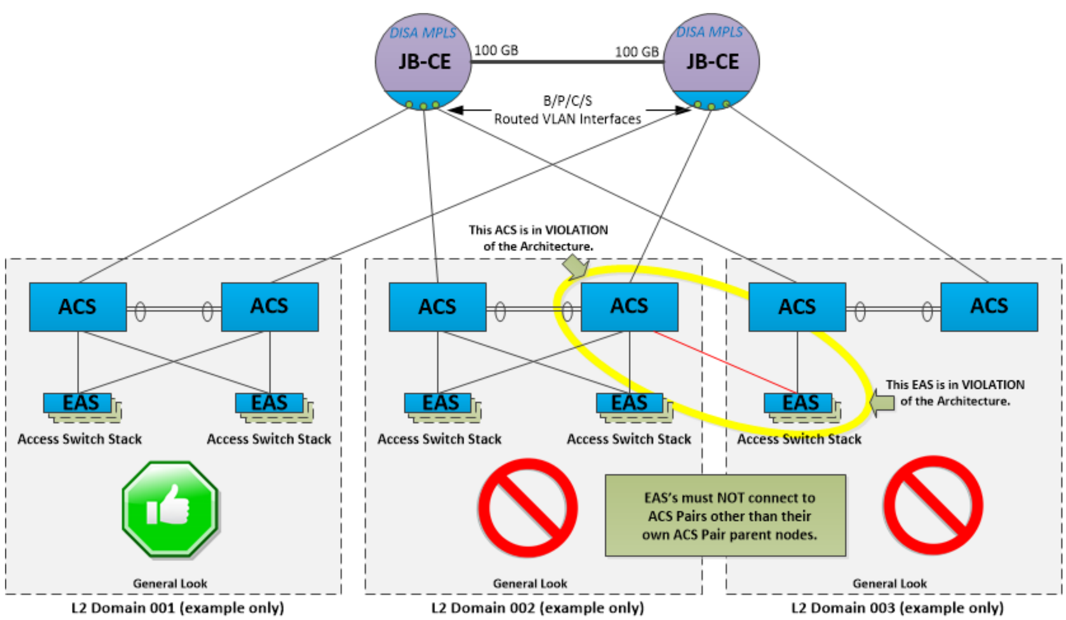


Diagram 3.3.2 – Example of what NOT to do (EAS Uplinks to Different L2 Domain ACS Switches)

Please note that by design an ACS pair defines a Layer-2 Area. Each Layer-2 Area hosts Base Unique as well as L2 Area unique VLANS. If an EAS is connected to more than a single Layer-2 Area, not only will spanning tree not function as planned but you may also run into problems with VLAN tag reuse. As a point of architecture, EAS devices are only authorized to connect to ADS and ACS devices within their own Layer-2 Area.

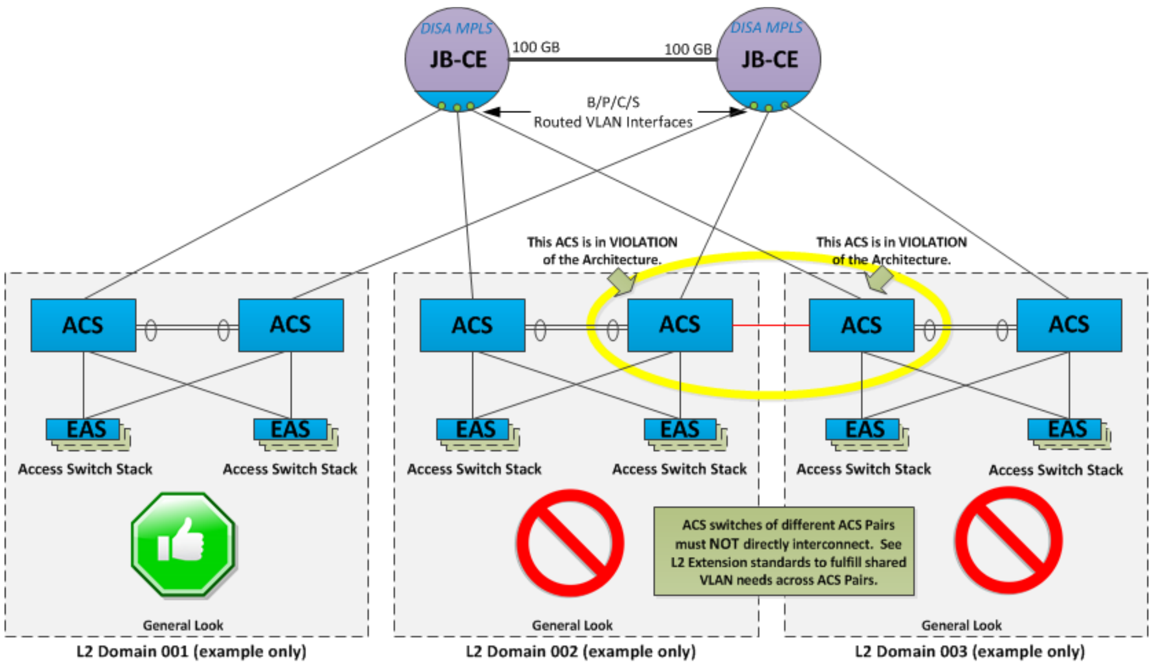


Diagram 3.3.3 – Example of what NOT to do (ACS Interconnectivity)

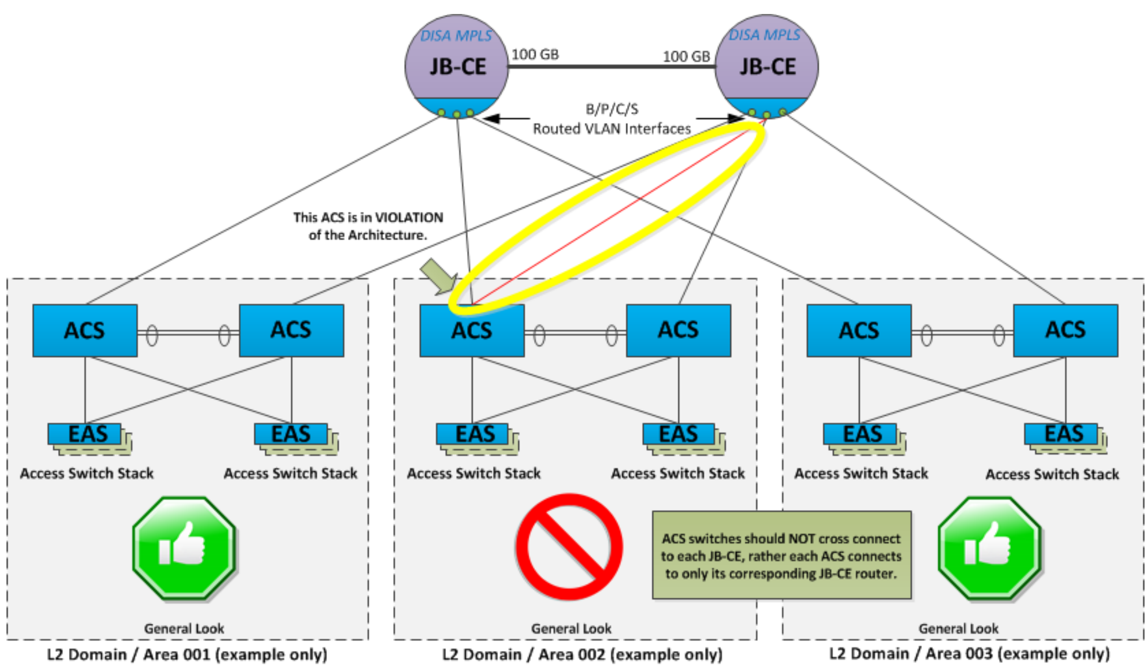
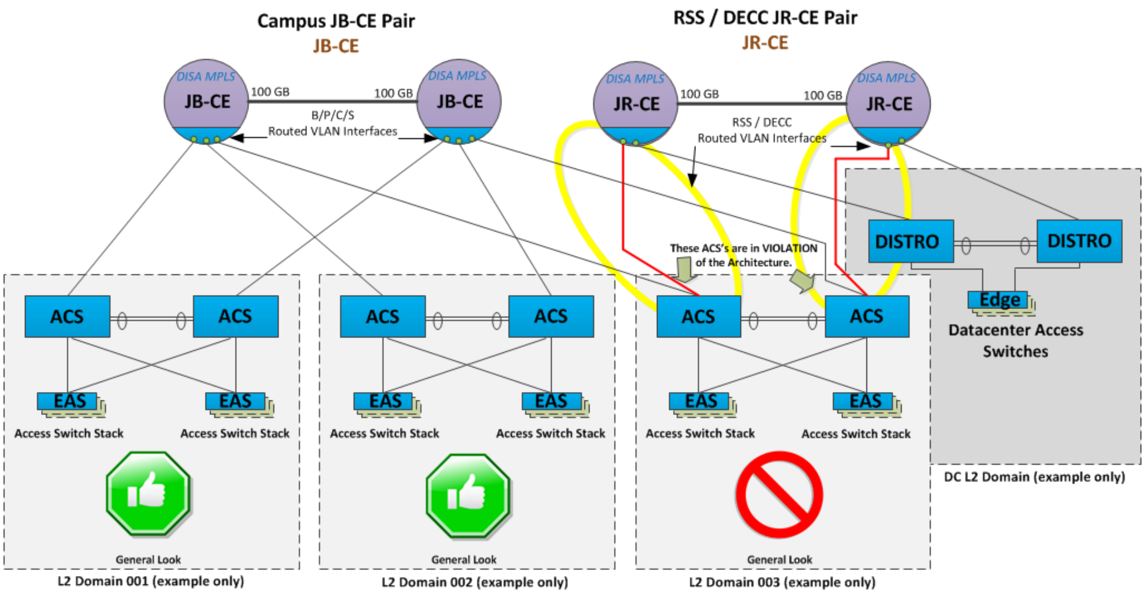


Diagram 3.3.4 – Example of what NOT to do (ACS to JB-CE Interconnectivity)



In JRSS locations, it is possible to have both a JB-CE (serving the user community) and a JR-CE (serving the JRSS stack and regional datacenters) on the same B/P/C/S. In these rare cases, be sure to connect your ACS devices to the JB-CE set of MPLS routers.

Diagram 3.3.5 – Example of what NOT to do (ACS Dual CE Uplink Connectivity)

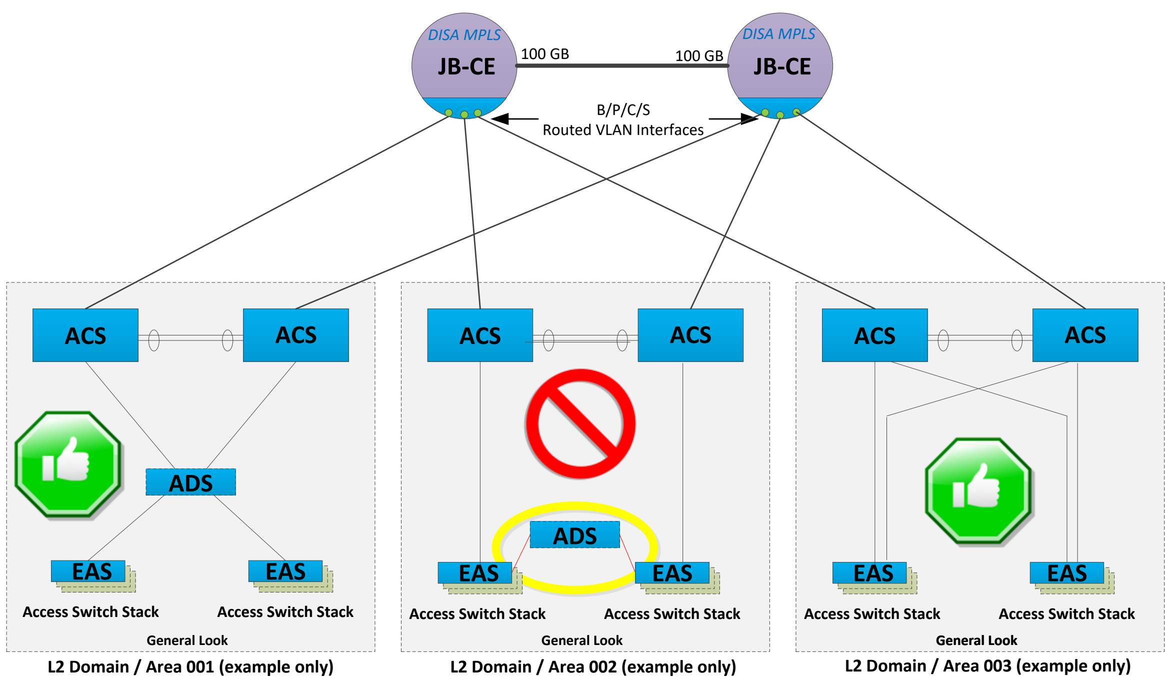


Diagram 3.3.6 – Example of what NOT to do (ADS used as a fiber coupler)

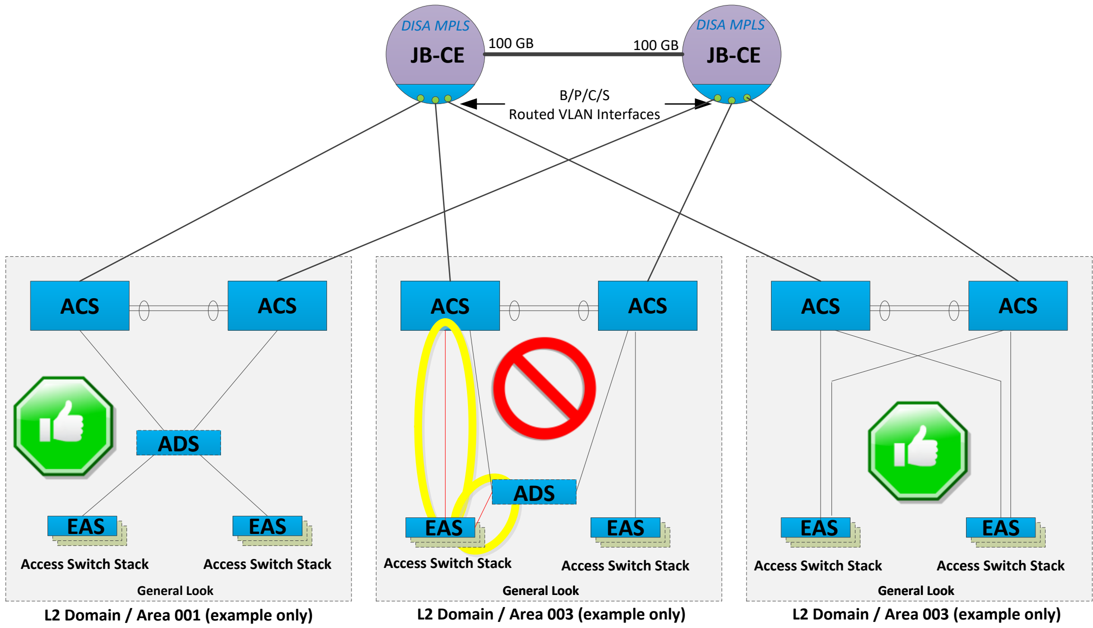


Diagram 3.3.7 – Example of what NOT to do (EAS uplinked via ACS and ADS at the same time)

The architecture does not legitimize "daisy chaining" of EAS devices, however it does recognize that in some locations there exists only limited amount of fiber to designated service areas. In the event that there is no other option but to daisy chain, the deviation needs to be clearly documented and provided in Army's case to NETCOM and in Air Force's case to AFSPC for awareness / risk management.

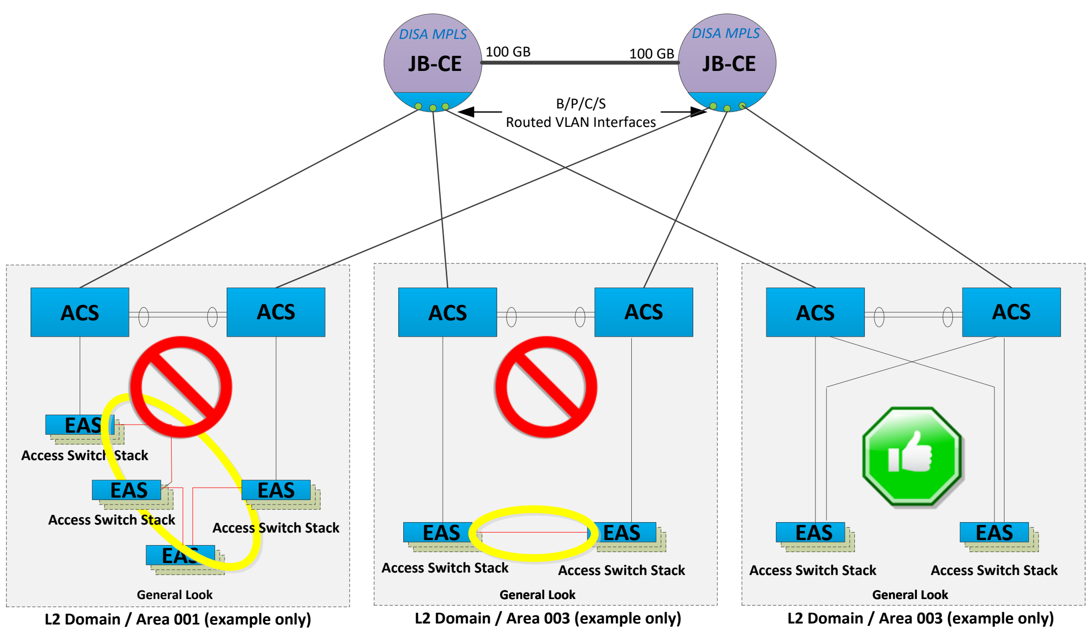


Diagram 3.3.8 – Example of what NOT to do (Directly interconnecting or daisy chaining EAS')

The Do's and Don'ts – A Summary

Permissible:

- EAS' will only connect to a single L2 Domain/Area (via an ADS or an ACS Pair)[1].
- EAS' will rely on switch-stacking as much as possible, up to 7 edge switches maximum per stack (creating virtual edge switches) to minimize uplink requirements.
    - Please note that if a vendor's equipment has a lower limit, that limit MUST be followed.
    - Please note that stacking implies the use of built in stacking ports connected to the stacking backplane.
- Remote EAS devices apart from being part of a remote ACS Pair, may only connect to the Base Executive Agent Collection System and rely on link encryption due to being external to the local ICAN (unless waivered).
- ADS devices do not directly connect to the JB-CE routers; they connect only to ACS devices.
    - Proper ADS/ACS relationship: within the economy of the same L2 Domain.
- ADS devices may only connect to a single L2 Domain/Area (ACS Pair)[1].
- Switch interconnections should be made utilizing Single Mode Fiber where possible. If not possible, it should be documented for remediation and reviewed annually.
- ACS' uplink to its upstream/host set of JB-CE routers.
- The two ACS aggregation switches within a given L2 Domain will be interconnected using link aggregation of two physical ports on each switch (from different modules)[2].
    - If a single service module exists in an ACS due to scarce port density requirements in a L2 Domain/Area, a non-redundant inter-connection to the peer ACS is sufficient because the ACS that loses its service module is automatically removed as a valid path to the Next Hope Redundant Routing Protocol defined Route Point.
- ACS switches from different L2 Domains/Areas shall NEVER be directly interconnected[3].
    - Each ACS within an L2 Domain/Area will only uplink to its designated JB-CE router and port; no cross-connects from an ACS to both JB-CE routers in the JB-CE pair.
- Please note that the Base Executive Agent can co-host ACS devices in the same facility as the JB-CE routers to aggregate ADS and EAS devices that are wired to the JB-CE rooms.
- The L2BS device singly attaches to any ACS Pair, this lack of connectivity redundancy is done on purpose/by design (it is not a flaw), in order to prevent accidental loops between L2 Domains[4].
- The L2BS device connects to each JB-CE router, using sub-interfaces on those physical ports on the JB-CE to service L2BS VLANs. L2BS VLANS are not bridged from ACS' to JB-CE routers.
- ***Using Remote ACS Pairs (with redundant uplinks)***
    - A remote site may be treated as a single or as multiple ACS pairs (independent L2 Domain), uplinked to a parent B/P/C/S JB-CE set of routers for route points (L3 Services).

[1] Please note that each ACS Pair represents the top aggregation nodes of all the underlying subordinate switches (potentially multiple ADS and a host of EAS devices) and are treated as belonging to the same Layer 2 Domain.
[2] This is required to avoid "*split brain routing*" with whichever Next Hope Redundancy Protocol is used.
[3] Each ACS Pair is representative of its own Layer 2 Domain/Area within the overall Campus Network, this allows for reuse of VLAN tags within a campus and it significantly stabilizes STP. Please see the Layer 2 Extension section to see how the architecture gracefully handles the interconnectivity of nodes that must be bridged across the campus.
[4] This solution was developed and successfully used by the Pentagon to limit cross-wedge broadcast storms/service impacts. The likelihood that a misconfiguration will occur over time is high, minimizing the chance of loops between areas manages both the risk and the potential impacts.

  - These remote sites should only connect to a single parent site pair of JB-CE routers and it needs to be able to do so in a redundant manner (at least a pair of fiber to each JB-CE router).
  - Because the local and the parent sites are geographically distinct, please remember that links leaving a B/P/C/S must be encrypted according to transport security laws currently in place.
- ***Tying in a collection of remote switches from a site without redundant uplinks***
  - In the event that a remote site is comprised of a collection of switches but its aggregation switch only has a single pair of fiber to a parent site, it will need to aggregate to the Base Executive Collection System of the parent site.
  - The aggregation switch in this instance is treated as an ADS switch.
  - Because the local and the parent sites are geographically distinct, please remember that links leaving a B/P/C/S must be encrypted according to transport security laws currently in place.
- ***Using a Remote Edge Access Switch (EAS)***
  - For truly small sites, a single EAS (which may be stacked) may be used. Remote EAS switches aggregate to the Base Executive Collection System of the parent site.
  - Because the local and the parent sites are geographically distinct, please remember that links leaving a B/P/C/S must be encrypted according to transport security laws currently in place.
- In all cases, when an exception is taken to the model architecture it must be documented and reported to the Base Executive Agent Director so that an exception report may be recorded and filed for risk assessment and possible future correction.

Impermissible:

- EAS' connected to two L2 Domain/Areas creating a L2 bridge.
- EAS' utilizing any other ports other than stacking ports to stack switches.
- ADS devices directly connected to the JB-CE routers rather than ACS devices.
- ADS devices bridging L2 areas.
- L2BS switches not redundantly uplinked to JB-CE routers.
- L2BS switches redundantly paired to both ACS' in an ACS Pair to bridge between L2 areas.
- The connection of ACS's in two separate L2 areas should occur only through a L2BS or via the JB-CE routers. They should not be directly interconnected.
- The "daisy-chaining" of EAS' is not acceptable. EAS' should only connect to ACS or ADS infrastructure.

4.0 CONFIGURATION SPECIFICATIONS

4.1 Device Host Naming Standards

The DISN-EN maximum recommended switching platform hostname length is up to 63 characters[5]. Although the same format is applied to all switches within the enterprise environment (EAS, ADS, ACS, and L2BS's), their format between (Edge/Distribution) vs. (Core) slightly varies to account for the relevant information needed by operations. Please note that the JB-CE hostname will adopt elements of the switch hostname format explained below, making all campus network devices follow a similar logic in naming convention.

The Site Identification (ID) is a globally unique, alpha/numeric shorthand for the name of the B/P/C/S[6]. The globally unique Site-IDs are DISA assigned (to include identifying Site-IDs for non DISN resident sites), currently handled and issued through the Joint Migration Team a subcomponent of DISA Global (DISA's JMT in support of infrastructure migration and modernization). See section 4.1.1 for more details on Side-ID Abbreviation.

The hostname relies on human-friendly naming down to the room level. More specific location information may be included in the executive banner to help locate a device. The fully qualified location of the device therefore is accessible upon login, via the executive banner[7] (see below) as well as the Simple Network Management Protocol (SNMP) System Location variables stored on the local switch in question.

```
banner exec ^C

!************ IMPORTANT DEVICE NOTES *******************!
!                                                       !
! Device Make/Model:             Cisco Nexus 7000       !
!                                                       !
! Parent/Uplink Pair:            JB-CE Pair 01          !
!                                                       !
! Building#:                     1342                   !
!                                                       !
! Room#:                         Server Room            !
!                                                       !
! Location:                      Row A/Rack 1           !
!                                                       !
! Additional Info:               *custom information    !
!                                                       !
!************ IMPORTANT DEVICE NOTES *******************!

^C
```

```
banner exec ^C

!************ IMPORTANT DEVICE NOTES *******************!
!                                                       !
! Device Make/Model:             Cisco Catalyst 3750-X  !
!                                                       !
! Parent/Uplink Pair:            ACS Pair 01            !
!                                                       !
! Building#:                     1200                   !
!                                                       !
! Room#:                         1543B                  !
!                                                       !
! Location:                      Row C/Rack 12          !
!                                                       !
! Additional Info:               *custom information    !
!                                                       !
!************ IMPORTANT DEVICE NOTES *******************!

^C
```

```
snmp-server location Ft. Hood,TX/South West Region

snmp-server contact JNOSC 800-255-5252
```

Diagram 4.1 – Banner and SNMP Diagram

[5] Contemporary switches can have hostnames up to 255 characters, however DISN-EN device hostnames shall not exceed 63 characters in length.
[6] Keeping the name humanly recognizable offers operations personnel clarity while effecting changes and troubleshooting.
[7] This banner is not displayed until AFTER login, thus not revealing any data to those without login credentials.

4.1.1 Site-ID Abbreviations

Within the ICAN, Site-ID abbreviations (a.k.a Base abbreviations) are included in L2 Device Hostnames and VLAN names. Site-IDs are comprised of five characters, the first three being specific to the Site Name and the last two designating geographic locations on the globe. These same exact Site-ID Abbreviations are also employed systematically on the Customer Edge Routers, Collection Routers, VRF Names under certain conditions and other Site Related Active IT configurations or device hostnames. Using ONE agreed-upon globally unique Site-ID enables enterprise operators to quickly find all site related items through this shared identifier. A list of pre-defined and approved Enterprise Site-ID abbreviations is available through DISA Global's JMT maintained JMS secure wiki site (https://wiki.jmn.mil) accessible only from within the secure Joint Management Network (JMN) which can be reached through secure VPN (access must be requested/approved through the JMT); you may inquire about or submit a change/new Site-ID request by contacting the JMT/SMT or the JMS Team.

Remote devices, such as ACS, ADS and EAS utilize their own sites names even though they are serviced by a parent B/P/C/S, in order to provide clarity on where the device is physically located.

4.1.2 EAS (Access Switch) Hostnames

Number of Characters Reserved: *up to 63 characters, with dash delimiters between data fields*

			Example:	c3750x-a01-hootx-4197-arnsth-0103-1s3						Up to 63 Characters					
Dynamic Length	*1*	*1*	*2*	*1*	*up to 5*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*1*	*1*	*1*
Vendor/Model	***Spacer***	***Function***	***L2 Area***	***Spacer***	***Site ID***	***Spacer***	***Building***	***Spacer***	***Room***	***Spacer***	***RackRow / Rack***	***Spacer***	***Device Instance***	***Stack Designator***	***# of Switches in Stack***
Alpha/Numeric	-	A (Access)	01	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	None	-	1	s	1
Alpha/Numeric	-	A (Access)	02	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	Desk	-	2	s	2
Alpha/Numeric	-	A (Access)	03	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	0102 (RR01.R02)	-	3	s	3
Alpha/Numeric	-	A (Access)	04	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	...	-	4	s	4
Alpha/Numeric	-	A (Access)	...	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	...	-	...	...	...
Alpha/Numeric	-	A (Access)	99	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	..	-	9	s	7

Diagram 4.1.2 – EAS Hostnames

Hostname indicates:

c3750x - Identifies the vendor (C-Cisco) and the model number (3750x) of the device.
a01 - Specifies that the EAS device is an (Access) Switch in L2 Domain 01.
hootx - Located in Fort Hood, TX (identifies specific B/P/C/S) derived from JMT Site-ID list
4197 - Identifies building 4197 as the host building for this device.
arnsth - Identifies room "arnsth" within the previously mentioned building as the location of the device
0103 - Identifies device in Rack Row 01 Rack 03.
1s3 - Identifies Device Instance 1, which is a stack made up of 3 switches. Please note that devices that are not comprised of a stack of switches are designated as 1s1.

The classification of the device, the theater it belongs to and the region in which the device resides in will be captured in the Domain Name System (DNS) Suffix and therefore are not necessary in the device hostname itself.

4.1.3 ADS (Distribution Switch) Hostnames

Number of Characters Reserved: *up to 63 characters, with dash delimiters between data fields*

			Example:	c3750x-d01-hootx-1531-120b-0103-1						Up to 63 Characters			
Dynamic Length	*1*	*1*	*2*	*1*	*up to 5*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*1*
Vendor/Model	***Spacer***	***Function***	***L2 Area***	***Spacer***	***Site ID***	***Spacer***	***Building***	***Spacer***	***Room***	***Spacer***	***RackRow / Rack***	***Spacer***	***Device Instance***
Alpha/Numeric	-	D (Distribution)	01	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	None	-	1
Alpha/Numeric	-	D (Distribution)	02	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	Desk	-	2
Alpha/Numeric	-	D (Distribution)	03	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	0102 (RR01.R02)	-	3
Alpha/Numeric	-	D (Distribution)	04	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	…	-	4
Alpha/Numeric	-	D (Distribution)	…	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	…	-	…
Alpha/Numeric	-	D (Distribution)	99	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	..	-	9

Diagram 4.1.3 – ADS Hostnames

Hostname indicates:

c3750x - Identifies the vendor (C-Cisco) and the model number (3750x) of the device.
d01 - Specifies that the ADS device is a (Distribution) Switch in L2 Domain 01 on the local B/P/C/S.
hootx - Located in Fort Hood, TX (identifies specific B/P/C/S).
1531 - Identifies building 1531 as the host building for this device.
120B - Identifies room 120B within the previously mentioned building as the location of the device.
0103 - Device is in Rack Row 01 and Rack 03.
1 - Identifies Device Instance.

The classification of the device, the theater it belongs to and the region in which the device resides in will be captured in the DNS Suffix and therefore are not necessary in the device hostname itself.

4.1.4 ACS (Core Switch) Hostnames

Number of Characters Reserved: *up to 63 characters, with dash delimiters between data fields*

			Example:	c3750x-c03a-hootx-1733-130-0301-1						Up to 63 Characters				
Dynamic Length	*1*	*1*	*2*	*1*	*1*	*up to 5*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	*1*	*1*
Vendor/Model	***Spacer***	***Function***	***L2 Area***	***JB-CE Identifier***	***Spacer***	***Site ID***	***Spacer***	***Building***	***Spacer***	***Room***	***Spacer***	***RackRow / Rack***	***Spacer***	***Device Instance***
Alpha/Numeric	-	C (Core)	01	A	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	None	-	1
Alpha/Numeric	-	C (Core)	02	B	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	Desk	-	2
Alpha/Numeric	-	C (Core)	03	-	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	0102 (RR01.R02)	-	
Alpha/Numeric	-	C (Core)	04	-	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	…	-	
Alpha/Numeric	-	C (Core)	…	-	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	…	-	…
Alpha/Numeric	-	C (Core)	99	"A" or "B"	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	1	..	-	

Diagram 4.1.4 – ACS Hostname

Hostname indicates:

c3750x - Identifies the vendor (C-Cisco) and the model number (3750x) of the device.
c03a - Is an ACS device belonging to L2 Domain/Area *03, attaching to the campus JB-CE Router 1 (A).* The A & B designation is important in helping operators be clear on which device to terminate L2BS connections on and which devices host which set of VLANs as the active MST root.
hootx - Located in Fort Hood, TX (identifies specific B/P/C/S).
1733 - Identifies building 1733 as the host building for this device.
130 - Identifies room 130 within the previously mentioned building as the location of the device.
0301 - Device is in Rack Row 03 and Rack 01.
1 - Identifies Device Instance.

The classification of the device, the theater it belongs to and the region in which the device resides in will be captured in the DNS Suffix and therefore are not necessary in the device hostname itself.

4.1.5 L2BS (Layer 2 Bridging Switch) Hostnames

Number of Characters Reserved: *Up to 63 characters, with dash delimiters between data fields*

			Example:	c3750x-l2bs-hootx-1773-130-0103-1					Upto 63 Characters			
Dynamic Length	*1*	*1*	*1*	*up to 5*	*1*	*Dynamic Length*	*1*	*Dynamic Length*	1	*Dynamic Length*	*1*	1
Vendor/Model	***Spacer***	***Function***	***Spacer***	***Site ID***	***Spacer***	***Building***	***Spacer***	***Room***	***Spacer***	***RackRow / Rack***	***Spacer***	***Device Instance***
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	None	-	1
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	Desk	-	2
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	0102 (RR01.R02)	-	…
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	…	-	
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	…	-	
Alpha/Numeric	-	L2BS	-	Alpha/Numeric	-	Alpha/Numeric	-	Alpha/Numeric	-	..	-	

Diagram 4.1.5 – L2BS Hostname

Hostname indicates:

c3750x - Identifies the vendor (C-Cisco) and the model number (3750x) of the device.
l2bs - Signifies that this is the L2BS (Bridging Switch) for the Base.
hootx - Located in Fort Hood, TX (identifies specific B/P/C/S).
1733 - Identifies building 1733 as the host building for this device.
130 - Identifies room 130 within the previously mentioned building as the location of the device.
0103 - Device is in Rack Row 01 and Rack 03.
1 - Identifies Device Instance.

The classification of the device, the theater it belongs to and the region in which the device resides in will be captured in the DNS Suffix and therefore are not necessary in the device hostname itself.

4.2 Virtual Local Area Network Numbering Schema

The Institute for Electrical and Electronics Engineers (IEEE) 802.1Q networking standard enables support for VLANs and the multiplexing of said VLANs across shared physical links through VLAN tagging. The IEEE 802.1Q standard is applicable to both IPv4 and IPv6 and utilizes a 12-bit field specifying the VLAN to which the Ethernet frame belongs, providing for 4096 possible VLANs (0000-4095). The DISN-EN architecture lends itself to permitting local administrators to identify and utilize VLAN tags as either Base Unique *(tag is unique across the entire base)* and/or L2 Domain/Area Unique *(tag is unique only to the L2 Domain that is using it)*. B/P/C/S user populations may range from just a few personnel to anywhere over 70,000+ personnel, the architectural distinction is found in the number of L2 Domains/Areas used to service the final campus network.[8] As such, L2 Domains/Areas represent physically distinct networks that are interconnected by the JB-CE routers via a Layer 3 protocol. This permits the reuse of VLAN tags across different L2 Domains/Areas and the scalability needed to account for B/P/C/S of any size. The DISN-EN L2 Domain/Area architecture also relies on Multiple Spanning Tree (MST) regional groupings, with a single L2 Domain/Area belonging to a single region. Each region may have multiple MST instances as is required. This enables load balancing of VLAN traffic across the ACS devices and minimizes network re-convergence impacts. The VLAN Numbering Schema presented here accounts for this the architecture, by systematically creating the two MST domain VLAN tag groupings (i.e.: Air Force *MST Group-1* is 2000 - 2149 while Air Force *MST Group-2* is 3000 - 3149, see matrix below).

DISN-EN Participating Base Executive Agent Administrators will assign VLAN Tags for their particular tenants from within the range of VLANs provided. Please note that when the VLAN tags already in use do not line up with the matrix provided below, the existing VLAN numbers must be renumbered in order to be compliant with this artifact. Various management and provisioning tools utilize the VLAN Names (which includes the VLAN Tag), VLAN Matrix compliance is required for these tools to work effectively in the Enterprise. When VLAN Tagging standards are combined with the VLAN Naming standard (see within this document), a fully qualified identifier is created giving operations a simple but scalable way of managing VLANs on their campuses.

For convenience, please see the VLAN Numbering Schema and summary matrix below.

[8] Please note that Category C Sites (bases with less than 200 personnel rely on ADS' instead of ACS' for aggregation).

VLAN Range *(1-4096)*							
Start	End	# of VLANs	Context	MST-Group	For Use By	Description	
0000	0000	1	Reserved	MST-00	No-One	Switch Platform Internal Use Only - Not Seen / Usable for Assignment	Special
0001	0001	1	Reserved	MST-00	No-One	Switch Platform Default VLAN (DISA STIG - Do Not Use)	Special
1000	1000	1	Reserved	MST-00	Reserved for DISN-EN Unused Ports	Vlan for Disabled Ports	Special
1001	1001	1	Reserved	MST-00	Reserved for Native VLAN	Reserved for Native VLAN	Special
1002	1002	1	Reserved	MST-00	No-One	Vendor Specific / Compatibility Reservation: Default FDDI VLAN	Special
1003	1003	1	Reserved	MST-00	No-One	Vendor Specific / Compatibility Reservation: Default Token Ring TrCRF	Special
1004	1004	1	Reserved	MST-00	No-One	Vendor Specific / Compatibility Reservation: Default FDDI NET VLAN	Special
1005	1005	1	Reserved	MST-00	No-One	Vendor Specific / Compatibility Reservation: Default Token Ring TrBRF	Special
1006	1199	144	Reserved	MST-00	No-One	Vendor Specific / Compatibility Reservation: Internal Allocation	Special
1150	1199	50	Reserved	MST-00	No-One	Reserved - Unused	Special
4094	4095	2	Reserved	MST-00	No-One	Switch Platform Internal Use Only - Not Seen / Usable for Assignment	Special
0002	0049	48	Reserved	MST-01	***Reserved for Future***	Enterprise Admin - Reserved for Future Use	Enterprise
0050	0099	50	Base Unique	MST-01	ENT - Backbone	Backbone (Routing) Admin Svcs (Unique Across Base) **(HAIPE/CT)**	Enterprise
0100	0149	50	Base Unique	MST-01	ENT - Management	Management Services (Unique Across Base)	Enterprise
0150	0199	50	L2 Area Unique	MST-01	ENT - IP Phones	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0200	0249	50	L2 Area Unique	MST-02	ENT - IP Phones	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0250	0299	50	L2 Area Unique	MST-01	ENT - VTC, IPTV, AV	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0300	0349	50	L2 Area Unique	MST-01	ENT - VDI (Zero Clients)	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0350	0399	50	L2 Area Unique	MST-01	ENT - MGMT WIFI and Wireless (Cell, LMR, WIDS)	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0400	0449	50	L2 Area Unique	MST-01	ENT - Base Security	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0450	0499	50	L2 Area Unique	MST-01	ENT - IPN MGMT (ILO, HYPERVISOR) & IT SECURITY	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Enterprise
0500	0549	50	Reserved	MST-02	***Reserved for Future***	Base Executive Agent - Reserved for Net Admin	Base Exec
0550	0599	50	Base Unique	MST-02	BEA - Net Admin	Backbone (Routing) Admin Svcs (Unique Across Base) **(HAIPE/CT)**	Base Exec
0600	0649	50	Base Unique	MST-02	BEA - Management	Management Services (Unique Across Base)	Base Exec
0650	0699	50	L2 Area Unique	MST-02	BEA - IP Phones	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0700	0749	50	L2 Area Unique	MST-01	BEA - IP Phones	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0750	0799	50	L2 Area Unique	MST-02	BEA - VTC, IPTV, AV	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0800	0849	50	L2 Area Unique	MST-02	BEA - VDI (Zero Clients)	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0850	0899	50	L2 Area Unique	MST-02	BEA - MGMT WIFI and Wireless (Cell, LMR, WIDS)	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0900	0949	50	L2 Area Unique	MST-02	BEA - Base Security	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
0950	0999	50	L2 Area Unique	MST-02	BEA - IPN MGMT (ILO, HYPERVISOR) & IT SECURITY	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Base Exec
1200	1249	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1250	1299	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1300	1349	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1350	1399	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1400	1449	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1450	1499	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1500	1549	50	Reserved	MST-01	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1550	1599	50	Reserved	MST-01	***Gov Guests (US Civilians)***	Reserved for Future Use	Future / Federal & Civ
1600	1649	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1650	1699	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1700	1749	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1750	1799	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1800	1849	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1850	1899	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1900	1949	50	Reserved	MST-02	***Reserved for Future***	Reserved for Future Use	Future / Federal & Civ
1950	1999	50	Reserved	MST-02	***Military Guests (Coalition Partners)***	Reserved for Future Use	Future / Federal & Civ
2000	2099	100	L2 Area Unique	MST-01	***Air-Force Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2100	2149	50	L2 Area Unique	MST-01	***Air-Force Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2150	2199	50	L2 Area Unique	MST-01	***Army Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2200	2299	100	L2 Area Unique	MST-01	***Army Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2300	2399	100	L2 Area Unique	MST-01	***Marine Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2400	2449	50	L2 Area Unique	MST-01	***Marine Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2450	2499	50	L2 Area Unique	MST-01	***Navy Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2500	2599	100	L2 Area Unique	MST-01	***Navy Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2600	2699	100	L2 Area Unique	MST-01	***Other Service/Agency Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2700	2799	100	L2 Area Unique	MST-01	***Medical Command Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2800	2899	100	L2 Area Unique	MST-01	***Joint Command Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2900	2909	10	L2 Area Unique	MST-01	Army Reserves Tenants	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
2910	2999	90	L2 Area Unique	MST-01	***MISC Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3000	3099	100	L2 Area Unique	MST-02	***Air-Force Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3100	3149	50	L2 Area Unique	MST-02	***Air-Force Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3150	3199	50	L2 Area Unique	MST-02	***Army Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3200	3299	100	L2 Area Unique	MST-02	***Army Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3300	3399	100	L2 Area Unique	MST-02	***Marine Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3400	3449	50	L2 Area Unique	MST-02	***Marine Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3450	3499	50	L2 Area Unique	MST-02	***Navy Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3500	3599	100	L2 Area Unique	MST-02	***Navy Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3600	3699	100	L2 Area Unique	MST-02	***Other Service/Agency Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3700	3799	100	L2 Area Unique	MST-02	***Medical Command Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3800	3899	100	L2 Area Unique	MST-02	***Joint Command Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3900	3909	10	L2 Area Unique	MST-02	Army Reserves Tenants	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
3910	3999	90	L2 Area Unique	MST-02	***MISC Tenants***	Re-Usable per L2 Area (Use with as many L2 Area as is Necessary)	Tenant VLAN Services
4000	4093	94	Base Unique	MST-15	***L2 Extensions***	L2 Extentions (by Request and Approval Only, Unique Across Base)	
	# of VLAN Tags Accounted for:	**4,096**					

	Tenant	MST-00	MST-01	MST-02	MST-15	# of VLANs:	Theoretical Number of Users ***Supported by SINGLE L2 Domain / Area***
	Reserved	0,1,4094-4095	N/A	N/A	N/A	4	800
Enterprise	ENT - Net Admin	N/A	0002-0099	N/A	N/A	98	10,000
Enterprise	ENT - Management	N/A	0100-0149	N/A	N/A	50	10,000
Enterprise	ENT - IP Phones	N/A	0150-0199	0200-0249	N/A	100	10,000
Enterprise	ENT - VTC, IPTV, AV	N/A	0250-0299	N/A	N/A	50	10,000
Enterprise	ENT - VDI (Zero Clients)	N/A	0300-0349	N/A	N/A	50	10,000
Enterprise	ENT - MGMT WIFI and Wireless (Cell, LMR, WIDS)	N/A	0350-0399	N/A	N/A	50	10,000
Enterprise	ENT - Base Security	N/A	0400-0449	N/A	N/A	50	10,000
Enterprise	ENT - IPN MGMT (ILO, HYPERVISOR) & IT SECURITY	N/A	0450-0499	N/A	N/A	50	10,000
Base Exec Agent	BEA - Net Admin	N/A	N/A	0500-0599	N/A	100	10,000
Base Exec Agent	BEA - Management	N/A	N/A	0600-0649	N/A	50	10,000
Base Exec Agent	BEA - IP Phones	N/A	0700-0749	0650-0699	N/A	100	10,000
Base Exec Agent	BEA - VTC, IPTV, AV	N/A	N/A	0750-0799	N/A	50	10,000
Base Exec Agent	BEA - VDI (Zero Clients)	N/A	N/A	0800-0849	N/A	50	10,000
Base Exec Agent	BEA - MGMT WIFI and Wireless (Cell, LMR, WIDS)	N/A	N/A	0850-0899	N/A	50	10,000
Base Exec Agent	BEA - Base Security	N/A	N/A	0900-0949	N/A	50	10,000
Base Exec Agent	BEA - IPN MGMT (ILO, HYPERVISOR) & IT SECURITY	N/A	N/A	0950-0999	N/A	50	10,000
	Special Purpose / Reserved	1000-1199	N/A	N/A	N/A	200	40,000
	Reserved for Future Use	N/A	1200-1549	1649-1949	N/A	700	10,000
	Gov Guests (US Civilians)	N/A	1550-1599	N/A	N/A	50	10,000
	Military Guests (Coalition Partners)	N/A	1950-1999	N/A	N/A	50	10,000
	Air-Force Tenants	N/A	2000-2149	3000-3149	N/A	300	160,000
	Army Tenants	N/A	2150-2299	3150-3299	N/A	300	60,000
	Marine Tenants	N/A	2300-2449	3300-3449	N/A	300	60,000
	Navy Tenants	N/A	2450-2599	3450-3599	N/A	300	60,000
	Other Service/Agency Tenants	N/A	2600-2699	3600-3699	N/A	200	40,000
	Medical Command Tenants	N/A	2700-2799	3700-3799	N/A	200	40,000
	Joint Command Tenants	N/A	2800-2899	3800-3899	N/A	200	40,000
	Army Reserves Tenants	N/A	2900-2909	3900-3909	N/A	20	5,000
	MISC Tenants	N/A	2910-2999	3910-3999	N/A	180	36,000
	L2 Extenstions	N/A	N/A	N/A	4000-4093	94	18,800

4.3 Virtual Local Area Network Naming Schema

The DISN-EN VLAN Naming schema starts with a letter for compatibility through the use of the VLAN Designator followed by the VLAN tag, the L2 Domain Area and the parent VRF-Name or Canonical Name (CNAME). Please note that for compatibility, the maximum number of characters supported in a VLAN name is 32.

Example:	v2155.01.army-hootx-usr-tr

1	4	1	2	1	4	1	upto 5	1	3	1	3
VLAN Designator	**VLAN-ID**		**L2 Area ID**		**VRF Name or CNAME (suggestion below)**						
v	0001	.	00	.	**Agency**		**Site-ID**		**Function**		**Zone**
v	0002	.	01	.	Alpha	-	Alpha Numberic	-	usr	-	tr
v	0003	.	02	.	Alpha	-	Alpha Numberic	-	ptr	-	dmz
	…	.	…	.	Alpha	-	Alpha Numberic	-	mgt	-	mgt
v	4094	.	99	.	…	-	…		…	-	…

Diagram 4.3 – VLAN Name

An example VLAN Name would be: **v2155.01.army-hootx-usr-tr**
Delineated version: Not needed, VLAN-Name inherently delineated.

VLAN Name indicates:

-----
v2155 - VLAN 2155.
01 - Identifies Servicing L2 Domain/Area 01.
<VRF Name or CNAME>
army - Service Agency, specifically ARMY.
hootx - Fort Hood, TX (uses JMT provided Site-ID).
usr - Specifies Function is that of servicing users.
tr - Specifies that this is in at trusted Security Zone.

VLAN Functions:

Predefined Functions:	
ADM	Admin Network
EDU	Training Network
MGT	Management Network
MSC	Miscellaneous Network
P2P	Point to Point Network
PRT	Printer Network
QNT	Quaranteen Network
SPN	Remote Span Network
SRV	Server Network
TEL	VoIP / Telephone Network
USR	User Network
VID	Video Network

Security Zone Identifiers:

Zone ID	***Security Zone Description***
tr	Trusted Zone
dmz	Demilitarized Zone
mgt	Management

SA Departments:

Service Agencies are comprised of many service departments. The Base Executive Agent has the freedom to define within five characters the naming of their service departments. Each department usually has a shorthand name they are already known by; it is recommended that the existing/known shorthand be adapted to the extent possible.

SA-Identifier:

Predefined Service Agencies:	
ARMY	US Army
NAVY	US Navy
USMC	US Marine Corps
AF	US Air Force
OSD	Office of the Secretary of Defense
JS	Joint Staff
MED	Medical
MISC	Miscellaneous
GST	Guest

The up-to four letter abbreviations for the known Service Agencies are identified in the table to the left. If a new/other Service Agency identifier is needed that is not already listed, the Base Executive Agents may add four letter identifiers to fit their network's needs.

4.4 Base Executive Agent VLAN Subnet Guidelines

The Base Executive Agent DISN-EN architecture subnet guideline specified in this section is applicable to user, printer and server VLANs *(non-applicable to routed backbone and network administrative subnets).* Please note that users, printers and servers will not share broadcast domains by design,

Bit Mask	***Total IP Addresses***	***Reservations***		***Tenant Usable***
		Network_ID to Network_ID+9	***Broadcast***	
/23	512	10	1	501
/24	256	10	1	245
/25	128	10	1	117
/26	64	10	1	53
/27	32	10	1	21
/28	16	10	1	5

but rather each must have its separate broadcast domain (VLAN)[9]. All VLANs except for the L2 Extension range of VLANs are to be kept local within a single L2 Domain/Area; they are not to be bridged across to other L2 Domains/Areas[10]. All user, printer and server VLANs/subnets are comprised of *"Admin Reserved"* and *"Tenant Usable"* IP addresses; where the "*Admin Reserved*" IP address range provides reservations for VRRP (3 addresses), troubleshooting (1 address), user experience monitoring (1 address), broadcast address (1 address) and 4 additional addresses reserved for future/unknown requirements. The "*Tenant Usable*" IP addresses may be assigned directly to whichever category of service it is providing on the network (users, printers and servers).

4.5 Access Switch, Port Configuration Standards

The DISN-EN EAS devices that directly connect to tenant devices like printers, desktops, phones and multimedia nodes are, as the name implies, access switches.

The following standards apply to the configuration of access switches:

4.5.1 Access Ports: (Downlinks)

- Safeguards:
  - Unused ports will be placed in VLAN 1000 "DISN-EN.UNUSED".
  - Unused ports will be administratively disabled/shut down. *Caveat:* Unused ports may be left enabled if **all** of the following conditions are met:
    - Unused ports are placed in VLAN 1000 "DISN-EN.UNUSED."
    - The ICAN has an implemented 802.1X solution.
    - The Base Executive Agent signs off on the deviation of this standard acknowledging the 802.1X solution provides an acceptable risk remediation.
  - All tenant facing ports will have a *Broadcast Guard* in place – Limit Ingress and Egress Broadcast traffic on the physical port up to 5% of the total link bandwidth capacity[11].
  - All tenant facing ports will have a *Multicast Guard* in place – Limit Ingress traffic on the physical port up to 5% of the total bandwidth capacity [12].
  - All tenant facing ports will have Dynamic Host Configuration Protocol (*DHCP)* Snooping in place – explicitly identifying the trusted uplink ports from which authorized DHCP Offer and Acknowledgement packets should source[13]. (STIG JIE-ICAN-061)
  - All tenant facing ports will be configured with Spanning Tree Protocol (STP) in such a way that ports immediately start in the forwarding state and transitioning to an error state if a Bridge Protocol Data Units (BPDU) is received from the user device[14].
  - All tenant facing ports will be configured with STP BPDU Guard in place – which gracefully handles the BPDU error state by simply shutting down the edge port if a BPDU is received. (STIG JIE-ICAN-058)

[9] This is to keep in step with the known JIE intent to keep said traffic types distinct, separate and measurable.
[10] Exception to this requires special approval and the use of specifically designated VLAN range utilization; please see the "*VLAN Numbering Schema*" section and the "*Base Executive Layer 2 Extension*" section of this document for further details.
[11] Prevents any one broadcast source to Denial of Service (DoS) the network, or network storm to DoS the workstation.
[12] Prevents any one multicast source to DoS the network.
[13] Monitors DHCP traffic, permits Discover and Request packets to source from tenants but only allows Offer and Acknowledgments packets to source from the uplinks.
[14] Enabled workstations and end stations to get access to the network immediately instead of waiting on the "listening" and "learning" states to complete.

        - The requirement for STP Loop Guard, which provides additional protection against loops, is covered by the STP BPDU Guard. (STIG JIE-ICAN-059)
    - All tenant facing ports will have IP Source Guard enabled – providing IP address filtering to prevent a malicious host from impersonating a legitimate host by impersonating the IP address. (STIG JIE-ICAN-062)
    - All tenant facing ports will be configured with BPDU error recovery enabled with a thirty second interval.
    - All tenant facing ports will have Unknown Unicast Flood Blocking (UUFB) enabled. (STIG JIE-ICAN-060)
    - All tenant facing ports will observe port security in the most secure way possible out of the following options:
        - **Weak Security:** Limit the number of Media Access Control (MAC) addresses to be learned from edge to a single MAC for ports servicing printers and three MAC addresses for ports servicing desktops with soft phones or ports with Physical IP Phones. A one-minute MAC inactivity aging setting should be enabled.
        - **Strong Security:** If a B/P/C/S or Headquarters has an existing implementation of 802.1x to authenticate its tenant devices, continue to use the existing 802.1x solution for VLANs servicing that VRF. With Base Executive Agent concurrence, tenants on the B/P/C/S not already in the VRF of the 802.1x implementation may be added to the implementation via approved MAC lists or be configured with the "*Weak Security*" noted above[15].
        - **Enterprise Security:** DISN-EN will provide an Enterprise 802.1x solution in the near future for all tenants on the DISN-EN; the availability of this solution has not yet been publicized and is not immediately available. Those joining the DISN-EN Architecture may claim this POAM to the known Security Technical Implementation Guide (STIG) finding without such solution in place.
- All tenant facing copper ports will be configured with Auto (Speed) and Auto (Duplex).
- All tenant facing fiber ports will be configured with Maximum (Speed) and Full (Duplex).
- All ports will conform to the DISN-EN Port Description Standard[16].
- Internet Group Management Protocol (IGMP) Snooping and Multicast Listener Discovery (MLD) Snooping will be enabled on the Access Switch to monitor and police multicast traffic. (STIG JIE-ICAN-039)
- All ports will have Unidirectional Link Detection (UDLD) enabled to protect against one-way connections. (STIG JIE-ICAN-053)
- All switches must have trunk links enabled statically; disable Dynamic Truck Protocol (DTP). (STIG JIE-ICAN-054)
- All switches must have Dynamic ARP Inspection (DAI) enabled on all user VLANs. DAI intercepts ARP requests and verifies that each of these packets has a valid IP-to-MAC address binding before updating the local ARP cache and forwarding the packet to the appropriate destination. (STIG JIE-ICAN-063)

4.5.2 ADS and ACS Facing Ports: (Uplinks)

- Uplink Ports will remain reserved if unused, they may not be utilized to service tenant devices.

[15] Not having 802.1x implemented on edge ports will create a category 2 STIG finding.
[16] Please see Port Description Standard in subsequent sub-section of this artifact.

- All uplink ports will conform to the DISN-EN Port Description Standard[17].
- All switches at the access layer must have an uplink to at least two separate nodes at the distribution layer. (STIG JIE-ICAN-052)

4.5.3 Management Port: (Console and MGMT IP Address)

- Single Switch Virtual Interface (SVI) per Edge Access Switch (EAS) will be configured for Management[18].
- Authentication, Authorization and Accountability (AAA) utilizing Radius and/or TACACS+ will be configured.
- Proxy Address Resolution Protocol (ARP) will be disabled.
- IP Redirects will be disabled.
- Telnet will be disabled, Secure Shell (SSHv2) will be enforced.
- Network Time Protocol (NTP) will be configured with authentication.
- System Logging (SYSLOG) will be configured.
- Simple Network Management Protocol version 3 (SNMPv3) will be configured with different community strings for READ and WRITE capabilities.
- Virtual Teletype (VTY) Lines will be secured with an Access Control List.
- Self-Signed certificates will be utilized until a Non-Person Entity (NPE) Public Key Infrastructure (PKI) is established to sign device certificates.

4.6 Distribution Switch, Port Configuration Standards

The DISN-EN ACS and ADS devices that directly connect to EAS devices are considered distribution switches. The following standards apply to the configuration of distribution switches:

4.6.1 EAS Facing Ports: (Downlinks)

- Safeguards:
  - Unused ports will be placed in VLAN 1000 "DISN-EN.UNUSED"
  - Unused ports will be administratively disabled/shut down.
  - All EAS facing ports will have a Broadcast Guard in place – Limit Ingress and Egress Broadcast traffic on the physical port up to 5% of the total link bandwidth capacity[19].
  - All EAS facing ports will be configured with STP. All EAS facing ports will be configured root bridge protection in all Spanning Tree types supported on the device.
    - Some sort of Root Guard should be a standard in MST mode if MST is the chosen Spanning Tree type. (STIG JIE-ICAN-057)
- All ports will conform to the DISN-EN Port Description Standard[20].
- All ports will have UDLD enabled to protect against one-way connections. (STIG JIE-ICAN-053)
- All switches must have trunk links enabled statically; disable DTP. (STIG JIE-ICAN-054)
- The assignment of EAS devices to distribution switch module/ports will, to the extent possible, be assigned in a symmetrical manner between the switches within an ACS pair of switches servicing EAS switches (i.e.: switch1.module2.port 13 and switch1.module3.port13 used to service same EAS switch… symmetrical port assignments).

[17] Please see Port Description Standard in subsequent sub-section of this artifact.
[18] In the event that an EAS is comprised of multiple physical switches, a single shared logical interface is used.
[19] Prevents any one broadcast source to DoS the network.
[20] Please see Port Description Standard in subsequent sub-section of this artifact.

4.6.2 ACS and JB-CE Facing Ports: (Crosslinks/Uplinks)

- Crosslinks/Uplink Ports will remain reserved if not used.
- Crosslinks between the two ACS switches in a given L2 Domain/Area will utilize 802.1AD (link aggregation whenever possible).
- All uplink ports will conform to the DISN-EN Port Description Standard[21].

4.6.3 Management Port: (Console and MGMT IP Address)

- ACS devices will use a physical out-of-band management port for management when available on ACS devices. ADS and EAS devices will use a single SVI for management, if the ACS cannot support a physical out-of-band management port it can also use a single SVI for management.
- AAA/Radius will be configured.
- Proxy ARP will be disabled.
- IP Redirects will be disabled.
- Telnet will be disabled, SSHv2 will be enforced.
- NTP will be configured with authentication.
- VTY Lines will be secured with an Access Control List.
- Self-Signed certificates will be utilized until a NPE PKI is established to sign device certificates.

4.7 Layer 2 Extensions

The DISN-EN architecture by design **disallows** spanning VLANs across distinct L2 Domains/Areas (VLANs should always rely on routing to cross L2 Domains/Areas) via the JB-CE devices. Furthermore, Layer 2 extensions shall not be implemented post-ICAN modernization without documented approval of the Base Executive Agent and only in cases where a L3VPN solution is not able to resolve the needs of the environment. It is recognized, however, that for some unique and legacy applications, the ability to share the same broadcast domain across an entire campus or across more than a single L2 Domain (ACS Pair), a safe means of enablement is required. As such, the DISN-EN prescribes, on a restrictive and temporary basis, a controlled and loop-free mechanism to address the rare instances in which extending VLANs across multiple L2 Domains/Areas (single broadcast domain across multiple L2 Domains/Areas) is required. The following requirements are applicable to extending VLANs onto more than one single L2 Domain/Area within a base:

1. The Base Executive Agent must approve Layer 2 Extensions across campus (between L2 Domains).
2. The VLAN that is to be extended must be assigned from the base-unique range of VLANs[22].
3. Although the implementation of L2BS is optional, if a need exists on campus to span a particular VLAN across more than a single L2 Domain/Area, a L2BS must then be implemented to enable the capability to extend VLANs across L2 Domains in a controlled/safe manner.
   a. Single "*L2BS*" device is installed per base (L2BS port speeds = ACS uplink speeds).
   b. The "*L2BS*" device is to be physically installed at either one of the JB-CE locations on base.
   c. The "*L2BS*" device is to be SINGLY attached to an L2 Domain, landing on ACS device "A".
   d. The only VLANs that are trunked through the "*L2BS*" are the active L2 Extension VLANs.
   e. The root bridge of the L2 Extension VLANs must be the L2BS "*L2BS*".
   f. The "*L2BS*" connects to both JB-CE routers for the routing of VLANS in range 4000-4093.
4. L2 Extension VLANs are not to be trunked on direct ACS to JB-CE interconnect links.

[21] Please see Port Description Standard in subsequent sub-section of this artifact.
[22] See VLAN Numbering Matrix for Reserved L2 Extension VLAN Ranges (4000-4094).

5. L2 Extension VLANs are trunked from the L2BS touch-point ACS switch in a given L2 Domain/Area to its area peer ACS switch and downstream ADS and/or EAS device.

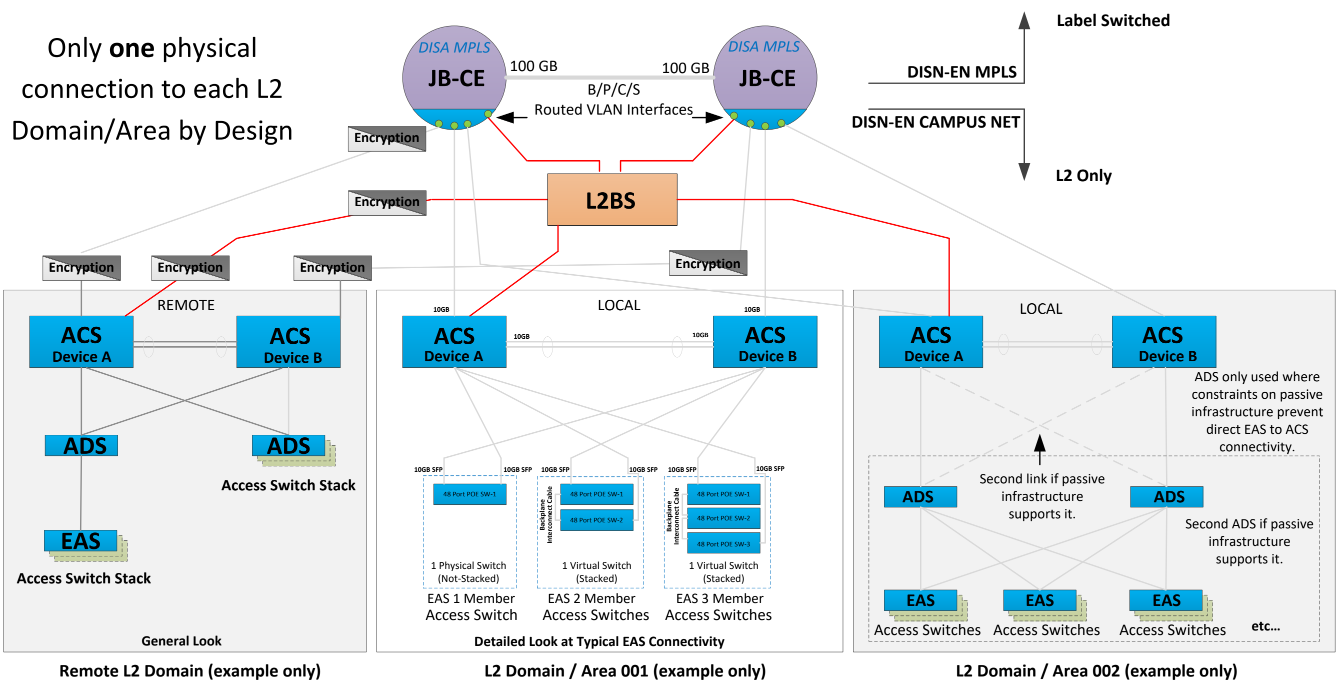


Diagram 4.7.1 – L2 Extension Physical Connectivity Drawing

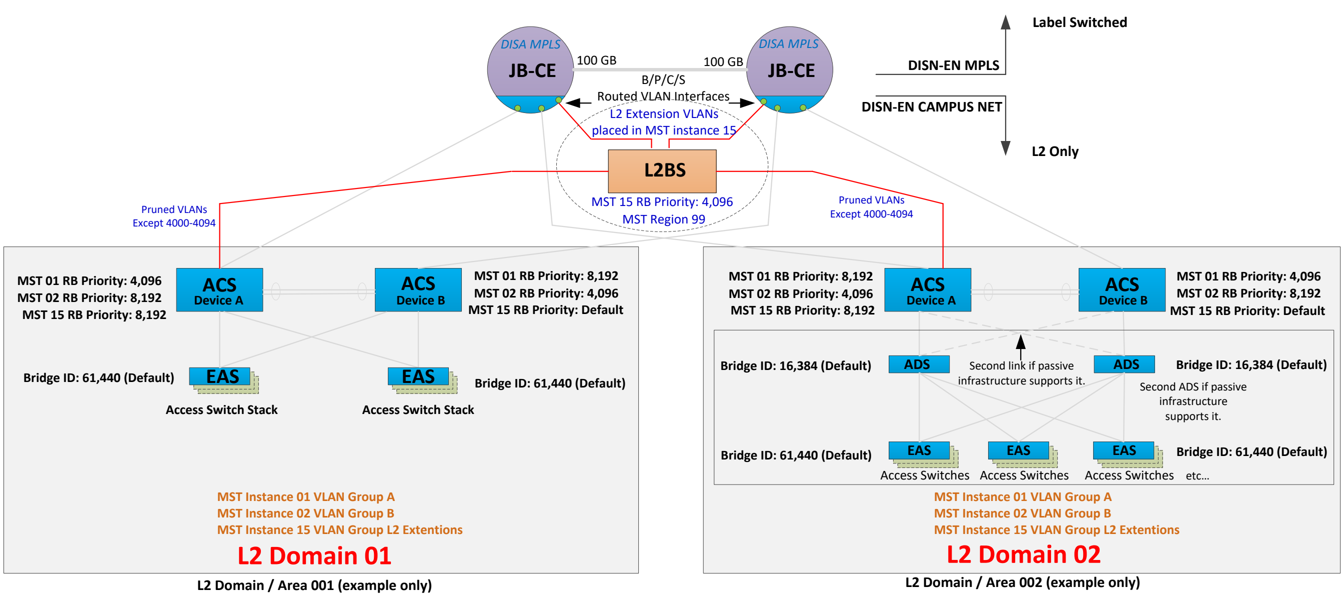


Diagram 4.7.2 – L2 Extension Logical Connectivity Drawing

4.8 Data Link Layer Encryption Standards

In unique cases, a need may arise for some enterprise tenants to hide their VLAN data from all other tenants within the campus by encrypting their payload across the B/P/C/S. Traditional means of accomplishing this goal, like using VPN concentrators or IPsec capable routers to secure traffic across the ICAN, remain valid solutions. However, the DISN-EN architecture provides an additional means of accomplishing that goal via 802.1ae Media Access Control Security (MACsec).

The DISN-EN network inherently supports 802.1ae at the ACS and ADS levels. To enable MACsec from EAS to EAS (one building to another building somewhere else on base), the EAS devices themselves must be 802.1ae capable and enabled. The DISN-EN EAS devices are capable, but to enable 802.1ae MACsec, a hardware encryption module may be necessary. The MACsec technology provides Federal Information Processing Standard (FIPS)-140-2 approved GCM-AES-128 encryption (Galois/Counter Mode of Advanced Encryption Standard cipher with a 128-bit key).

4.9 Root Bridge/Spanning Tree Standards

The DISN-EN architecture relies on the specific configuration of STP throughout the L2 Domain/Areas Network infrastructure in order to ensure the network maintains stability while providing redundancy through multiple paths with loop control.

- The DISN-EN will employ Multiple Spanning Tree Protocol 802.1s.[23]
- The DISN-EN will utilizes manual load balancing in the form of MST Load Balancing. MST Load Balancing enables each MST instance to take a different network path, thereby distributing traffic load across multiple paths. Each L2 Domain is serviced by two ACS devices; ACS 01 is the MST root for half of the VLANs while ACS 02 is the MST root for the other half of the VLANs.
- Within any given L2 Domain/Area, the MST region name must match along with its configuration revision and version number.
    - MST Region Names will be issued by the Base Executive Agent with cognizance that Region 99 is reserved for the L2BS L2 Domain.
- The DISN-EN will utilize a single vendor campus network (layer 2 devices) solution to ensure interoperability; this will be observed preferably across the entire base but at a minimum by L2 Domain/Area.
- The Layer 2 topology must be protected against splitting into smaller independent Layer 2 networks, this is accomplished by using link aggregation between the two ACS devices in the L2 Domain to ensure a unified switch fabric. The redundant links should be provisioned off of two distinct modules and as a single channel under 802.1ad Link Aggregation Control Protocol (LACP).
- All VLANs with an L2 Domain/Area will be trunked to every switch in the L2 Domain, however access port provisioning is done only on an as needed basis.
- The native VLAN on all trunk ports must meet one of the following requirements:
    - Frames in the native VLAN must be untagged.
    - The native VLAN must be a unique VLAN that is not used for any access port[24].
- MST will be configured on all switches within a particular L2 Domain/Area. (STIG JIE-ICAN-056)
- The default routing point for a STP Instance (produced via VRRP in a routed-redundant configuration) must always be aligned to the root bridge of the MST Instance – even in the presence of multiple link failures.

[23] http://www.ieee802.org/1/pages/802.1s.html
[24] Please note that VLAN ID 1001 has been identified as the Native VLAN for DISN-EN Campus Networks.

4.9.1 MST Regions

Each ACS Pair represents an L2 Domain. Each L2 Domain must have its own MST Region. For example, a B/P/C/S with three sets of ACS Pair switches would have three L2 Domains and three distinct MST Regions plus one more for authorized L2 Extensions via the L2BS (L2 Domain).

- Region 1 L2 Domain/Area 01.
- Region 2 L2 Domain/Area 02.
- Region 3 L2 Domain/Area 03, and so forth.
- Region 99 Used for the L2BS.[25]

4.9.2 MST Groups

Each Layer 2 Broadcast Domain will host multiple MST Groups:

- MST00 VLANs 1, 1000-1199 (VLANS with Special Exclusions).
- MST01 VLANs 2-199, 250-499, 700-749, 1200-1599 and 2000-2999.
- MST02 VLANs 200-249, 500-699, 750-999, 1600-1999 and 3000-3999.
- MST15 VLANs 4000-4093 (For use with L2 Extensions).

Please note that only 16 MST groups (instances) are supported on some vendor devices. Since the only MST group that crosses more than a single L2 Domain is MST Region 99, for the sake of scalability all other MST Region IDs can repeat in distinctly separate L2 Domains without conflict.

4.9.3 MST Naming Standard:

Example:	hootx.01.1377-7462.army.mil					
Up to 5	*1*	*2*	*Dynamic length*	*1*	*Dynamic Length*	Dynamic Length
Site-ID	***Spacer***	***MST Region ID***	***ACS Bldg A***	***Spacer***	***ACS Bldg B***	***Base Executive Agent***
Alpha/Numberic	.	01	Alpha/Numeric	-	Alpha/Numeric	Army.mil
Alpha/Numberic	.	02	Alpha/Numeric	-	Alpha/Numeric	AF.mil
Alpha/Numberic	.	03	Alpha/Numeric	-	Alpha/Numeric	Navy.mil
…	…	..	…	…	…	
Alpha/Numberic	.	15	Alpha/Numeric	-	Alpha/Numeric	

Diagram 4.9.3 – MST Naming

Hostname indicates:

hootx: Specifies unique Site-ID which may be up to 5 characters in length.
01: Specifies MST Region 01 as the context.
ACS Bldg A: ACS Pair Leg 1 (First ACS in Pair).
ACS Bldg B: ACS Pair Leg 2 (Second ACS in Pair).
Base Executive Agent: Identifies who the Base Executive Agent is for the Network.

[25] MST Region 99 uses Instance 15

4.9.4 Root Bridges

Root Bridges are configured per MST instance and per MST region. Valid Bridge Priorities are in the range 0 – 61,440 in multiples of 4096. Priority 0 must be left open for an event where the root bridge needs to be changed to restore service. The default priority of EAS devices should be 61,440 and the default priority of ADS devices should be set to 16,384.

- ACS Root Bridge Priority : 4,096
- ACS Backup Root Bridge Priority : 8,192
- L2BS Extension Root Bridge Priority : 4,096

4.10 Port Description Standards

The DISN-EN Port Descriptions are limited to 255 alpha/numeric characters. Typical information captured in the port description includes details such as port data or from/to port data to facilitate applications like E-911 as well as troubleshooting by operations personnel. Please see below for the recommended port descriptions. ICAN administrators may append additional data to the port descriptions; the following is the minimum requirement.

User Facing Port Descriptions (Ports on EAS')

						Example:	USR ; BLDG1023 ; RM3509 ; LOCNA ; DROP38-4D					
3	*3*		*Variable*	*3*		Variable	*1*		*Variable*	*1*		*Variable*
Function	***Spacer***	***Bldg Prefix***	***Bldg***	***Spacer***	***Room Prefix***	***Room***	***Spacer***	***Location Prefix***	***Specific Location***	***Spacer***	***Face Plate Prefix***	***Face Plate***
USR	;	BLDG	Alpha/Numeric	;	RM	Alpha/Numeric	;	LOC	NA (Not Applicable)	;	DROP	Alpha/Numeric
PRT	;	BLDG	Alpha/Numeric	;	RM	Alpha/Numeric	;	LOC	Alpha/Numeric	;	DROP	Alpha/Numeric
TEL	;	BLDG	Alpha/Numeric	;	RM	Alpha/Numeric	;	LOC	Alpha/Numeric	;	DROP	Alpha/Numeric
VID	;	BLDG	Alpha/Numeric	;	RM	Alpha/Numeric	;	LOC	Alpha/Numeric	;	DROP	Alpha/Numeric

Backbone Uplink/Downlink Port Descriptions (Uplink ports on EAS', and ports on ADS and ACS devices)

	Example:	c3750x-d01-hootx-1531-120b-0103-1 ; IP xxx.xxx.xxx.xxx ; ge1/0/15			
	up to 63	*3*	*Variable*	*3*	*Variable*
	Device Hostname	***Spacer***	***Mgmt IP Address***	***Spacer***	***Port***
	Alpha/Numeric	;	Alpha / Numeric	;	Alpha / Numeric
	Alpha/Numeric	;	Alpha / Numeric	;	Alpha / Numeric
	Alpha/Numeric	;	Alpha / Numeric	;	Alpha / Numeric
	Alpha/Numeric	;	Alpha / Numeric	;	Alpha / Numeric

Diagram 4.10.1 – Port Description Naming

4.11 Telecommunications Room Patch Cable Labeling Standards

Telecommunications Room Patch Cables will be labeled in accordance with the Technical Criteria for the Installation Information Infrastructure Architecture document. At a minimum, identical labels will be placed on each end of the patch cable using machine printed labels that are self-laminating. Each label will include both the near and far end patching information as shown in Diagram 4.11, Patch Cable Labeling. The following schema applies to Telecommunications Room Patch Cable labels.

Example:	01.04.36.17B					
2	*1*	*2*	*1*	*Variable*	*1*	*Variable*
RackRow	***Spacer***	***Rack***	***Spacer***	***Rack Unit***	***Spacer***	***Port***
01	.	01	.	01	.	Alpha/Numeric
02	.	02	.	02	.	Alpha/Numeric
03	.	03	.	03	.	Alpha/Numeric
…	…	..	…	…	…	…
99	.	99	.	99	.	Alpha/Numeric

Diagram 4.11 – Patch Cable Labeling

4.12 ICAN Class of Service (CoS) Standards

The DISN-EN ICAN Architecture relies on Class of Service (CoS) across its L2 Domains/Areas in order to prioritize the delivery of traffic across the Campus L2 Infrastructure and to the JB-CE campus routers. Prioritization (priority queuing) is accomplished by leveraging the *User Priority* field in the 802.1q packet headers. The Base Executive Agent is responsible for ensuring that L2 Class of Service is deployed on all switch interfaces on campus.

4.12.1 Layer 2 Ingress CoS and DSCP Trust Model

Edge ports servicing Voice, Video, Management, Point-to-Point and Server type traffic will be configured to trust the Differentiated Service Code Point (DSCP) or CoS marking as follows:

- If the interface is an 802.1q tagged trunk (i.e. Voice Over Internet Protocol (VoIP) Phone with a PC behind it, Softphone… etc.), then the ingress switch shall be configured to trust the CoS markings.
- If the interface is an un-tagged access port, the switch should trust the DSCP marking.
- All inter-switch trunk links will be configured to trust the CoS markings.

All other types of ports are to be considered "*untrusted*" with no preferential treatment (best effort delivery). The ICAN may reclassify the trust level of individual ports in this category (i.e. Command and Control (C2) systems) based on need.

Vendor specific features may be exploited beyond those stated here as deemed appropriate per the ICAN, provided the above requirements are satisfied.

4.12.2 Layer 2 Egress Queuing

All interfaces shall be configured to provide a priority queue for CoS value 5 *(permits assignment of DSCP priority traffic to be treated as CoS value 5 precedence queuing)*. The priority queue shall have limited de-queuing (i.e.: set limit on how much traffic is permitted to be treated as priority traffic) to prevent starvation of other queues.

- Recommendation is to set limit at 10% but this number may be modified as needed by the ICAN.
- Vendor specific features may be exploited beyond those stated here as deemed appropriate per the ICAN, provided the above requirements are satisfied.

4.12.3 Other Considerations/Comments

In order to equip the ICAN implementation team with awareness of intent, as to how traffic will be treated at the JB-CE, it is worth noting that the JB-CE sub-interfaces will be configured with ingress classification based on deep packet inspection (DPI) (certain exceptions will exist). As the trust model is extended into the B/C/P/S, it is unnecessary to perform ingress classification on the JB-CE for VLANs servicing Voice, Video, Management, Point-To-Point, and Servers. The assumption is that in all of those cases the integrity of the CoS markings has been ensured within the L2 Domain/Area based on the aforementioned requirements (extended trust model). Therefore, DPI-based ingress classification will only be required on a subset of the configured VLAN sub-interfaces. Furthermore, the ingress classification policy may be tailored on specific VLANs to reduce the Computer Processing Unit (CPU) load and delay. For example, the guest VLAN may have a simple ingress policy that marks all guest traffic with the appropriate DSCP value to place this traffic in the scavenger or default class on the MPLS backbone. The JB-CE routers will map IP precedence to the MPLS Traffic Class bits to provide the Assured Forwarding Per-Hop-Behavior (PHB) across the backbone. The details of these ingress and egress classifiers and queues for the JB-CE are contained in the DISN EN MPLS standards document.

4.13 ICAN Device Management Standards

The DISN-EN Architecture specifies that EAS devices will be managed in-band, while L2BS, ADS and ACS devices will be managed in one of two ways; preferably via a semi-out of band (OOB) solution (detailed below) but if that is not feasible for whatever reason then at a minimum via an in-band solution. L2BS, ADS and ACS devices should use the preferred method of management (semi-OOB) to allow the Joint Network Command Center (JNCC) engineers to reach and access the critical infrastructure components of an ICAN even in case of switch operating system software failures. Ultimately all L2 Infrastructure Device management traffic is injected back into the survivable/redundant network (in-band via a dedicated set of Management VLANs) and mapped into the DISN-EN Management VRF at the JB-CE routers. The aforementioned Management VLANs will terminate into the management Layer 3 Virtual Private Network (L3VPN) on the JB-CE's, ensuring complete segmentation between management and user traffic. The Base Executive Agent Network personnel will determine the size and quantity of Management VLANs within each L2 Domain/Area. The details of the Management L3VPN architecture and how the JB-CE itself will be managed will be defined in the DISN-EN MPLS standards document. The Out of Band Management network is logically out of band, while in some cases it may use ports which are physically in-band. Additionally, the use of switched virtual interfaces (SVI) will utilize physically in-band ports to connect to a logically segregated out of band network.

Control plane policing should be utilized to help ensure out of band management access to devices. This will allow the preference and preemption of management traffic to a device which may be experiencing a duress situation with a high throughput of customer traffic or other bandwidth demands.

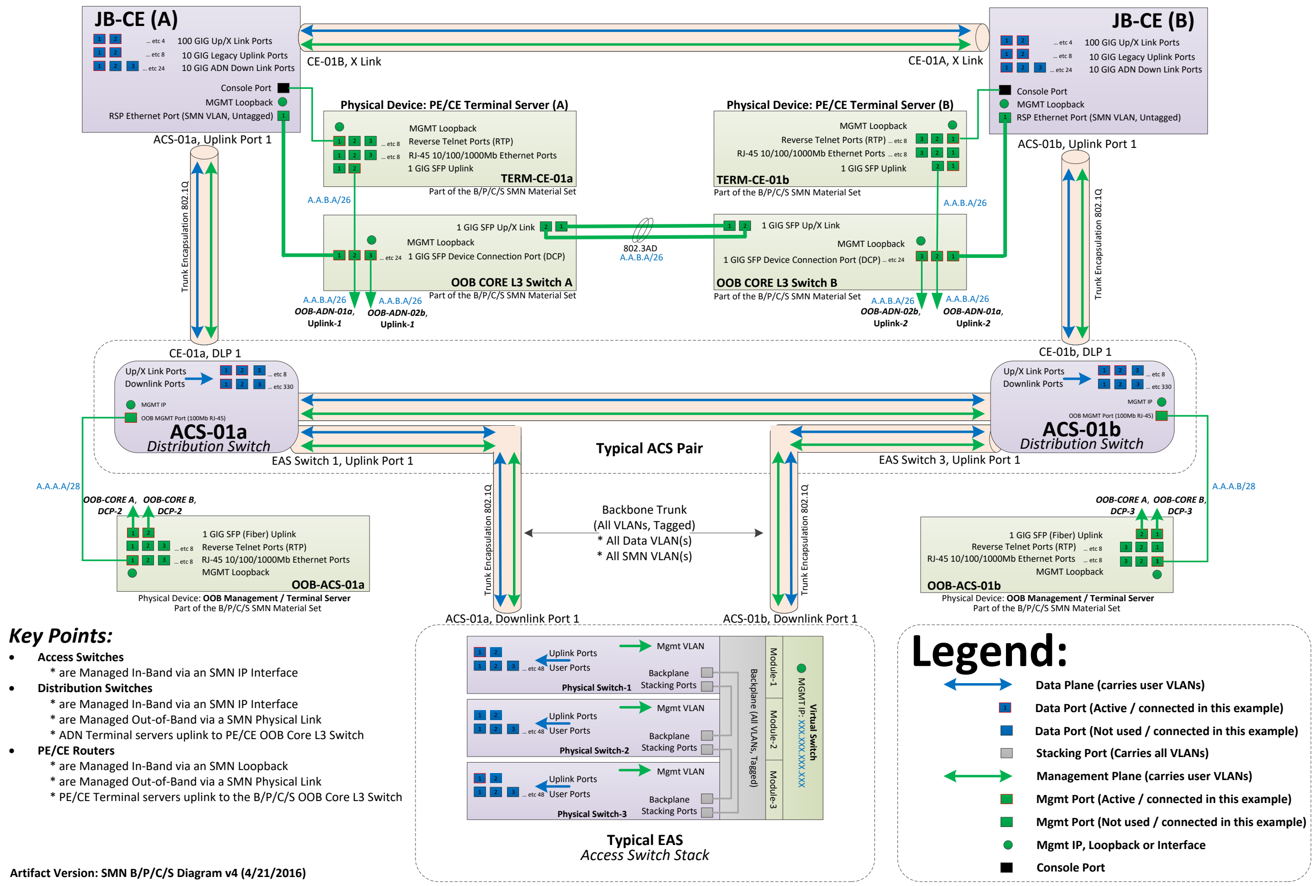


Diagram 4.13.1– Management Connectivity Drawing

Availability Reference Chart *(per user numbers)*

Availability %	Downtime per year	Downtime per month*	Downtime per week
90% ("one nine")	36.5 days	72 hours	16.8 hours
95%	18.25 days	36 hours	8.4 hours
97%	10.96 days	21.6 hours	5.04 hours
98%	7.30 days	14.4 hours	3.36 hours
99% ("two nines")	3.65 days	7.20 hours	1.68 hours
99.5%	1.83 days	3.60 hours	50.4 minutes
99.8%	17.52 hours	86.23 minutes	20.16 minutes
99.9% ("three nines")	8.76 hours	43.8 minutes	10.1 minutes
99.95%	4.38 hours	21.56 minutes	5.04 minutes
99.99% ("four nines")	52.56 minutes	4.32 minutes	1.01 minutes
99.999% ("five nines")	5.26 minutes	25.9 seconds	6.05 seconds
99.9999% ("six nines")	31.5 seconds	2.59 seconds	0.605 seconds
99.99999% ("seven nines")	3.15 seconds	0.259 seconds	0.0605 seconds

Target Architecture Availability (Excerpt from Unified Capabilities Requirements (UCR))

The ICAN LAN shall have four (4) nines of availability as its general target.

LAN REQUIREMENT ITEM	USER PRECEDENCE ORIGINATION AUTHORIZATION			
	FO/F	I/P	R	NOT MISSION CRITICAL
ASLAN High	R	P	P	P
ASLAN Medium	NP	R	P	P
Non-ASLAN	NP	NP	P	P
ASF	R	R	R	NR
Diversity	R	R	NR	NR
Redundancy	R	R	NR	NR
Battery Backup	8 hours	2 hours	NR	NR
Single Point of Failure User > 96 Allowed	No	No	Yes	Yes
GOS p=	0.0	0.0	0.0	Note 1
Availability	99.999	99.997	99.9	99.9

NOTE 1 GOS is discretionary and shall be determined by DoD Components.

LEGEND

ASF	Assured Services Features	I/P	IMMEDIATE/PRIORITY	P	Permitted
ASLAN	Assured Services LAN	LAN	Local Area Network	R	Required
FO/F	FLASH OVERRIDE/FLASH	NP	Not Permitted	R	ROUTINE
GOS	Grade of Service	NR	Not Required		

Diagram 4.13.2 – Availability

5.0 APPLICABLE DOCUMENTS

This section provides a list of documents that are applicable to the DISA-EN architecture. While every effort has been made to ensure the completeness of this list, other reference and standards documents may be applicable and therefore of high-value to the implementers.

5.1 Government Document

AR 25-2	Information Management Information Assurance
DISR	DoD IPv6 Standard Profiles, for IPv6 Capable Products, Version 2.0
JTA	DoD Joint Technical Architecture
JTA-A	Department of the Army Joint Technical Architecture
UCR 2013	DoD Unified Capabilities Requirements
UC APL	Unified Capabilities Approved Products List
STIG	Information Assurance Support Environment STIGS
Labeling	Technical Criteria for the Installation Information Infrastructure Architecture

5.2 Non-Government Document

National Electrical Code (NEC)
National Fire Protection Association (NFPA)
American National Standards Institute (ANSI)
Telecommunications Industry Association/Electronic Industries Association (TIA/EIA) 606
Administration Standards for the Telecommunications Infrastructure of Commercial Buildings (TIA/EIA) 569A

IEEE 802.1p Ethernet QoS
IEEE 802.1q VLAN Tagging
IEEE 802.1s Multiple Spanning Tree
IEEE 802.1x Port Security
IEEE 802.1ab Link Layer Discovery Protocol (LLDP)
IEEE 802.1ae MACsec
IEEE 802.3ad Link Aggregation
IEEE 802.3z 1GB Ethernet
IEEE 802.3ae 10GB Ethernet
IEEE 802.3at Power over Ethernet +

RFC 2475 Architecture for Differentiated Services
RFC 2474 DIFFSERV Field in the IPv4 and IPv6 Headers
RFC 1757 Remote Network Monitoring MIB
RFC 1907 Simple Network Management Protocol version 2
RFC 3411 Simple Network Management Protocol version 3
RFC 3246 Expedited Forwarding PHB
RFC 2597 Assured Forwarding PHB Group
RFC 2460 IPv6 Specification
RFC 2461 IPv6 Neighbor Discovery
RFC 2462 IPv6 Stateless Address Auto-Configuration
RFC 2463 ICMPv6
RFC 4291 IPv6 Addressing Architecture
RFC 3587 IPv6 Global Unicast Address Format
RFC 2375 IPv6 Multicast Address Assignment

RFC 2464 Transmission of IPv6 over Ethernet Networks
RFC 3596 DNS Support
RFC 2893 Transition Mechanisms for IPv6 Hosts and Routers
RFC 2328 OSPFv2 (with IPv4 Support Only)
RFC 5340 OSPFv3 (with IPv6 Support)
RFC 2338 Virtual Router Redundancy Protocol
RFC 1541 Dynamics Host Configuration Protocol
RFC 1654 Border Gateway Protocol version 4
RFC 1825 Security Architecture for the Internet Protocol
RFC 1828 IP Authentication using Keyed MD5
RFC 7426 Software-Defined Networking

STD-6 User Datagram Protocol
STD-7 Transmission Control Protocol
STD 13 Domain Name Systems
SDT 15 Simple Network Management Protocol

6.0 GLOSSARY/ACRONYMS

AAA - Authentication, Authorization and Accounting - a security architecture for distributed systems, which enables control over which users are allowed access to which services, and how much of the resources they have used

ACL – Access Control List

ACS – Area Core Switch

ADS – Area Distribution Switch – Intermediate aggregation switch that collects EAS devices before unlinking into the ACS devices.

AES – Advanced Encryption Standard

ANSI - American National Standards Institute

ARP - Address Resolution Protocol - a telecommunications protocol used for resolution of network layer addresses into link layer addresses.

ASLAN - Assured Services Local Area Network

Authenticator - Software or service running on a wireless access point or switch that manages the authentication process between the supplicant and authentication server.

Authentication server - Software or service that provides authentication services to the authenticator. Using the credentials provided by the supplicant, the authentication server controls whether the supplicant is authorized to access the services provided on the authentication server's protected network.

Joint Base Collection Router – A Joint Base Collection Router is a DISA owned/life-cycled set of routers and switches that enable circuit collection for circuits needing to uplink through the IPT-PE (inner core) routers for transport. The Joint Base Collection Router is managed through the JMS.

Base Executive Collection System – A set of Base Executive Service Agency Owned/life-cycled routers and switches working together to service the outer core (DISN-EN) circuit collection needs. The Base Executive Collection System is managed through the JMS through delegated credentials and attaches only to the JB-CE routers for upstream connectivity.

B/P/C/S – Base/Post/Camp/Station

BPDU - Bridge Protocol Data Units - data frames used by a bridge to exchange information about bridge IDs and root path costs.

CDP - Cisco Discovery Protocol - network protocol used to discover and share information between connected Cisco equipment.

CLI - Command Line Interface

CNAME – Canonical Name (domain name system record)

CoS – Class of Service

CPU – Computer Processing Unit

C2 – Command and Control

DAI – Dynamic ARP Inspection

DHCP - Dynamic Host Configuration Protocol - network protocol used to dynamically assign IP addresses to devices that are connected to a network.

DISN-EN - Defense Information Systems Network - Enterprise Network - DISA managed enterprise network architecture.

DoS - Denial of Service - making a machine or network resource unavailable to its intended users.

DNS – Domain Name System

DPI – Deep Packet Inspection

DSCP – Differentiated Service Code Point

DTP – Dynamic Link Protocol

DWDM - Dense Wavelength Division Multiplexing

EAP - Extensible Authentication Protocol - A generic authentication framework that supports many different types of authentication methods with different options, including Kerberos, public-key encryption, and one-time passwords.

EAPOL – EAP over Local Area Network

EAP-TLS - Extensible Authentication Protocol-Top Level Access

EAS – Edge Access Switch - End user network access switch

FIPS – Federal Information Processing Standard

HTTPS – Hyper Text Transfer Protocol Secure

I/O – Input/Output

ICAN - Installation Campus Area Network - the local LAN boundary at a base (the internal switches and desktops).

ID – Identification

IEEE – Institute for Electrical and Electronics Engineers

IGMP - Internet Group Management Protocol - a communications protocol used by hosts and adjacent routers on IP networks to establish multicast group memberships.

IP – Internet Protocol

IPN - Installation Processing Node - Serves a single DoD installation and local area (installations physically or logically behind the network boundary) with local services that cannot (technically or economically) be provided from a Core Data Center (CDC). Link

IPT–PE – IP Transport Provider Edge

IT – Information Technology

I3MP - Installation, Information, Infrastructure Modernization Program

JB-CE – Joint Base Customer Edge - Enterprise router which serves as the layer 3 access point onto the DISN-EN.

JIE – Joint Information Environment – DISN Future Target Architecture to which the DISN-EN Architecture is an enabler and facilitator.

JMN – Joint Management Network

JMS – Joint Management System

JNCC – Joint Network Command Center

JR-CE – Joint Regional Customer Edge

JRSS – Joint Regional Security Stack - Enterprise security architecture.

JTA – Joint Technical Architecture

LACP - Link Aggregation Control Protocol - provides a method to control the bundling of several physical ports together to form a single logical channel.

LLDP - Link Layer Discovery Protocol - vendor neutral protocol used to discover and share information between network devices.

LR – Long Reach

L2BS – Layer 2 Bridge Switch - Campus L2 Domain/Area interconnection switch that permits spanning select VLANs across multiple L2 Domains/Areas in a controlled and stable manner.

L3VPN – Layer 3 Virtual Private Network

MAB - MAC Authentication Bypass

MAC Address - Media Access Control Address - a unique identifier assigned to network interfaces for communications on the physical network segment.

MACsec – Media Access Control Security

MED – Media Endpoint Discovery

MDI/MDIX – Media Dependent Interface/Crossover

MD5 – Message Digest Algorithm

MLD - Multicast Listener Discovery

MPLS - Multi Protocol Label Switching - a mechanism in high-performance telecommunications networks that directs data from one network node to the next based on label lookups rather than network address lookups.

MST - Multiple Spanning Tree - a protocol which maps multiple VLANs into a spanning tree instance, with each instance having a spanning tree topology independent of other spanning tree instances.

MTR – Main Telecommunications Room

NEC – National Electrical Code

NFPA – National Fire Protection Association

NPE – Non-Person Entity

NTP - Network Time Protocol - a networking protocol for clock synchronization between computer systems over packet-switched, variable-latency data networks.

OEM – Original Equipment Manufacturer

OOB – Out of Band (Management Network)

OIR – Online Insertion and Removal

PC – Personal Computer

PHB – Per-Hop-Behavior

PKI – Public Key Infrastructure

P-Node – Provider nodes

POAM - Plan of Action and Milestones - a plan that describes specific measures to be taken to correct deficiencies found during a security control assessment.

POE – Power Over Ethernet

QoS – Quality of Service

RMON – Remote Monitoring

RU – Rack Unit

SDN – Software Defined Networks - A programmable networks approach that supports the separation of control and forwarding planes via standardized interfaces. IRTF RFC 7426

SFP – Small Form Factor Pluggable

SNMP - Single Network Management Protocol

SSH - Secure Shell - a cryptographic network protocol for secure data communication, remote command-line login, remote command execution, and other secure network services between two networked nodes.

STIG - Security Technical Implementation Guides - provide important reference guidelines and configuration standards for deploying approved technologies within the U.S. DoD.

STP - Spanning Tree Protocol - a network protocol that ensures a loop-free topology on a bridged Ethernet local area network.

Supplicant - Software or service running on a device that seeks access to a protected network.

SVI - Switch Virtual Interface

SYN – Synchronous

TACACS+ - Terminal Access Controller Access Control System +

TDR – Time Domain Reflectometry

TIA/EIA - Telecommunications Industry Association/Electronic Industries Association

TLA – Top Level Aggregation

TLS – Transport Layer Security

TCP – Transmission Control Protocol

TTBN - Top Tier Base Nodes - Top level network devices which serve as upstream aggregation points for lower level devices.

UC APL – Unified Capabilities Approved Products List

UCR – Unified Capabilities Requirements

UDLD – Unidirectional Link Detection

USCYBERCOM - United States Cyber Command

USG – United States Government

UUFB – Unknown Unicast Flood Blocking

VAC – Volts Alternating Current

VLAN - Virtual Local Area Network - a single layer-2 network broadcast domain.

VoIP – Voice Over Internet Protocol

VRF - Virtual Routing and Forwarding - a routing table instance, that can exist in one instance or multiple instances per each VPN on a Provider Edge (PE) router.

VRRP - Virtual Router Redundancy Protocol - network management protocol that is used to increase the availability of default gateway servicing hosts on the same subnet.

VTY - Virtual Teletype - command line interface that enables users to connect to a network device using the Telnet or SSH protocol.

WAN - Wide Area Network - A network that spans over a diverse geographical area.

Appendix

FOR THE

DISN Enterprise Network (DISN-EN)

Switch Device Requirements Specifications

FOR THE

INSTALLATION CAMPUS AREA NETWORK DESIGN & IMPLEMENTATION (ICAN-DI)

Artifact Version 1.09f FINAL
(5 December 2016)

For Official Use Only (FOUO)

7.0 APPENDIX

This section identifies the minimum **L2BS, ACS, ADS** and **EAS** device technical specifications to ensure that the architecture can be instantiated as architected through a known and common set of baseline technical capabilities. Adherence to this section enables architecture compatibility and sustainability.

7.1 Overarching Specifications and Standards Requirements (Applicable to all ICAN devices)

7.1.1 Technology Refresh:

- Equipment must be delivered from Original Equipment Manufacturer (OEM) with the latest Unified Capabilities Approved Products List (UC APL) approved technology (e.g. latest memory standard) and software/firmware revisions installed.

7.1.2 IPV6 Compliance

- All hardware and associated software must be IPv6 capable. An IPv6 capable system or product shall be capable of receiving, processing, transmitting and forwarding IPv6 packets and/or interfacing with other systems and protocols in a manner similar to that of IPv4. Specific criteria to be deemed IPv6 capable are:
    - Maintenance of interoperability with IPv4. Systems being developed, procured or acquired shall maintain interoperability with IPv4 systems/capabilities.
    - Systems must implement IPv4/IPv6 dual-stack and should also be built to determine which protocol layer to use depending on the destination host it is attempting to communicate with or establish a socket with.
    - Availability of IPv6 technical support for system development, implementation and management.

7.1.3 Manufacturing Requirements:

- Minimum of 8 years of OEM support before end of life (3 years manufacturing and 5 years support).
- Certified as genuine by manufacturer's representative time of receipt by government
- On DoD UC APL at time of award
- Must be Assured Services Local Area Network (ASLAN)-certified

7.1.4 IEEE Standards:

- IEEE 802.1AB LLDP.
- IEEE 802.1p Ethernet CoS.
- IEEE 802.1q VLAN Tagging.
- IEEE 802.1s Multiple Spanning Tree.
- IEEE 802.1x Port Security.
- IEEE 802.3z 1GB Ethernet.
- IEEE 802.3ad LACP.
- IEEE 802.3ae 10GB Ethernet (Small Form-factor Pluggable (SFP+) Long Reach (LR) optical transceivers).
- IEEE 802.1ae Media Access Control (MAC) Security.
- IEEE 802.3ba 40/100GB Ethernet (Backbone/Uplink Ports)

7.2 Hardware Requirements:

7.2.1 For Chassis Based Systems:

- Control plane and data forwarding modules shall be physically separated.
- Distributed forwarding capability shall be performed by the Input / Output (I/O) module.

7.2.2 For All Systems:

- Must be capable of non-blocking as specified in UCR-2013 section 7.2.1 for maximum performance levels for the specified minimum number of ports of each of the switch platforms as stated in their respective sections of this document. Ports in excess of the minimum required need not be non-blocking.
- Power supplies must be configured to maximum redundancy (N+1 at a minimum).
- All other internal components (fans, etc.) shall be redundant (N + M) and load sharing (where applicable) to maximum available. N + M redundancy is defined as followed: N is the number of active components required to operate a system without redundancy; M is the number of "spares" specified to provide the requisite level of system availability. All fans and power supply internal components, shall be field replaceable non-disruptively, and must support Online Insertion and Removal (OIR).
- Must include an out of band management interface to provide remote management and troubleshooting capabilities.

7.3 Software Requirements:

The acquired systems must be supported through software, documentation, licenses, version upgrades, patch upgrades, and help desk support for all equipment described above for a period of five years after the date of installation of the equipment.

- The software must be an included component of the equipment purchase.
- A single software license model that includes all required features.
- Capable of OpenFlow as an industry-standard Software Defined Network (SDN) protocol that facilitates communication between the forwarding plane of a network switch or router and the controller that enables network-wide flow control.
- Must support a minimum hostname length of 63 characters.

7.4 Network Management Standards:

- Command Line Interface (CLI).
- Dedicated out-of-band management port using 10/100 Mbps minimum.
- Dedicated out-of-band management port using CLI-based console port.
- Management via a single IP interface per device (a stack is treated as one device).
- SNMP Version 3 w/AES Encryption.
- IPv6 Management.
- SNMP MIB II.
- Remote Monitoring (RMON).
- NTP with Message Digest Algorithm (MD5) authentication.

7.5 Security Standards:

The security of ports subject to IA requirements is governed by DISA STIG guidelines. All switches connected to the DISN-EN are bound by these guidelines. For a complete listing of STIGs please refer to the Information Assurance Support Environment (IASE) DISA website, **http://iase.disa.mil/stigs/Pages/index.aspx**.

- Protection for Denial of Service protection against Transmission Control Protocol Synchronous (TCP SYN) or Smurf Attacks.
- Must support Access Control Lists (ACL) on Console, VTY, a per subnet and per VLAN (VLAN ACL (VACL) basis.
- AAA.
- RADIUS.
- TACACS+.
- Secure Shell (SSH v2).
- Secure Copy (SCP v2).
- Hyper Text Transfer Protocol Secure (HTTPS).
- Username/Password (Challenge and Response).
- Bi-level Access Mode (Standard and Privileged Level).

7.6 QoS (Quality of Service) Standards:

- RFC 2474 DIFFSERV Field in the IPv4 and IPv6 Headers.
- RFC 2475 An Architecture for Differentiated Services.
- RFC 3246 An Expedited Forwarding PHB.
- RFC 2597 Assured Forwarding PHB Group.

7.7 IPv6 Standards:

- RFC 2460 IPv6 Specification.
- RFC 2461 IPv6 Neighbor Discovery.
- RFC 2462 IPv6 Stateless Address Auto-Configuration.
- RFC 2463 ICMPv6.
- RFC 4291 IPv6 Addressing Architecture.
- RFC 3587 IPv6 Global Unicast Address Format.
- RFC 2375 IPv6 Multicast Address Assignments.
- RFC 2464 Transmission of IPv6 over Ethernet Networks.
- RFC 3596 DNS support.
- RFC 2893 Transition Mechanisms for IPv6 Hosts and Routers.

7.8 EAS (Edge Access Switch) Requirements

There are two types of EAS devices, the ones that support copper ports and ones that support fiber ports. The copper port switches may be one of two types, a traditional telecommunications rack mounted model and a small form factor model to be installed in special use considerations. The type of switches that will need to be installed will be dependent on the physical infrastructure available at a given site. All EAS must support the following requirements without the addition of hardware, software, firmware or licensing:

7.8.1 Standard Edge Access Switch

7.8.1.1 *Standard Edge Access Switch General System Requirements*

- One rack unit (RU) in height (1RU=1.75inch).
- Must be available to purchase in the following configurations:
  - Minimum Quantity of 48, 10/100/1000 BaseT copper user/downlink ports.
  - Minimum Quantity of 24 100/1000 Base-X SFP fiber user/downlink ports.
- Input voltage capable of both 120/220 Volts Alternating Current (VAC).
- Redundant power supplies (N+1 Redundancy).
- Hot-swappable power supplies.
- Redundant and Hot-swappable fans.
- Must support stacking (supporting a minimum of 240 ports in a stack). Stacking hardware must be included in proposed solution.
- Must maintain power redundancy, whether 110 VAC or 220 VAC, for control/data functions and simultaneous class-3 Power Over Ethernet (POE) on every client facing copper edge port in the event of a single power supply failure either via individual switch's power redundancy or power redundancy within a switch stack configuration.
- Must be able to change, remove or disable the default VLAN from trunks.
- Must support up to 4094 VLAN Tags.
- Must support at a minimum eight (8) queues for traffic prioritization.
- Must be able to assign traffic to queues.
  - By VLAN tags, types of traffic or DIFFSERV Code Point.
  - Set Quality of Service (QoS) by DIFFSERV Code Point based on type of traffic (e.g. VoIP).
- Must support IGMP Snooping.
- Must support minimum Jumbo Ethernet frame size of 9000 bytes.
- Capable of filtering spanning-tree BPDU.
- 802.1x - Port-Based Network Access Control supporting Extensible Authentication Protocol-Top Level Access (EAP-TLS) using DoD PKI Machine Identity Certificates.
  - 802.1x – MAC Authentication Bypass (MAB) for non-802.1x capable hosts.
- 802.1x - Multi-Domain support for separation of voice and data.
- 802.1x - Fail Open (for migrating to 802.1x).
- 802.1AB – Link Layer Discovery Protocol (LLDP)/LLDP-Media Endpoint Discovery (MED).
- Support IEEE 802.3ba 40/100GB Ethernet (Backbone/Uplink Ports) for future-proofing.

7.8.1.2 *Standard Edge Access Switch Uplink Ports:*

- Minimum of two 10GbE with 1GbE backwards compatibility.
- Capable of full line rate (e.g. 2 x 10GB/s Uplink) when fully loaded.
- Must be MACSec capable.

7.8.1.3 *Standard Edge Access Switch Downlink Port Requirements:*

All edge ports (user ports) must support the following without the addition of hardware, software, firmware or licensing:

- 48 ports of PoE+ (IEEE 802.3at-2009) with 25.5 watts of power/port.
- Spanning-tree BPDU detection and protection.
- Time Domain Reflectometry (TDR) diagnostic testing capability.
- Must support auto Media Dependent Interface/Crossover (MDI/MDIX).

- Must support Source Guard (STIG JIE-ICAN-061 and -062).
- Must support Unknown Unicast Flood Block (UUFB) (STIG JIE-ICAN-060).
- Must support Root Guard (STIG JIE-ICAN-057).

7.8.2 Small Form Factor EAS

7.8.2.1 *Small Form Factor EAS General System Requirements*

- One rack unit (RU) in height (1RU=1.75inch).
- Must be available to purchase in the following configuration:
  - Minimum Quantity of 12 10/100/1000 BaseT copper user/downlink ports.
  - Must be capable of full line rate speed on all 10/100/1000BaseT downlink ports.
- Input voltage capable of both 120/220 Volts Alternating Current (VAC).
- Must be able to change, remove or disable the default VLAN from trunks.
- Must support up to 4094 VLAN Tags.
- Must support at a minimum eight (8) queues for traffic prioritization.
- Must be able to assign traffic to queues.
  - By VLAN tags, types of traffic or DIFFSERV Code Point.
  - Set Quality of Service (QoS) by DIFFSERV Code Point based on type of traffic (e.g. VoIP).
- Must support IGMP Snooping.
- Must support minimum Jumbo Ethernet frame size of 9000 bytes.
- Capable of filtering spanning-tree BPDU.
- 802.1x - Port-Based Network Access Control supporting Extensible Authentication Protocol-Top Level Access (EAP-TLS) using DoD PKI Machine Identity Certificates.
  - 802.1x – MAC Authentication Bypass (MAB) for non-802.1x capable hosts.
- 802.1x - Multi-Domain support for separation of voice and data.
- 802.1x - Fail Open (for migrating to 802.1x).
- 802.1AB – Link Layer Discovery Protocol (LLDP)/LLDP-Media Endpoint Discovery (MED). Small Form Factor EAS Uplink Ports:
- Minimum of two 1GbE SFP or SFP+ based uplink ports.
- Capable of full line rate (e.g. 2 x 1GB/s Uplink) when fully loaded.

7.8.2.2 *Small Form Factor EAS Downlink Port Requirements:*

All edge ports (user ports) must support the following without the addition of hardware, software, firmware or licensing:

- Minimum four (4) ports of PoE+ (IEEE 802.3at-2009) with 25.5 watts of power/port.
- Spanning-tree BPDU detection and protection.
- Time Domain Reflectometry (TDR) diagnostic testing capability.
- Must support auto Media Dependent Interface/Crossover (MDI/MDIX).
- Must support Source Guard (STIG JIE-ICAN-061 and -062).
- Must support Unknown Unicast Flood Block (UUFB) (STIG JIE-ICAN-060).
- Must support Root Guard (STIG JIE-ICAN-057).

7.9 ADS (Area Distribution Switch) Requirements

ADS will be used by the United States Government (USG) to aggregate multiple switches geographically dispersed in a building prior to reducing the number of uplinks needed between an ACS and EAS. A common use may be in a main telecommunication room (MTR) to aggregate the traffic from switches in secondary TRs. Where outside plant infrastructure exists, the preferred solution is to utilize the MTR as a fiber distribution node (FDN) to connect EAS' back to the appropriate ACS. An ADS capability should only be used when the outside plant cannot support EAS direct uplinks to the ACS.

All ADS must support the following requirements without the addition of hardware, software, firmware or licensing:

7.9.1 General Requirements

- Must be ASLAN-Certified.
- Input voltage capable of both 120/220 VAC or 48 VDC.
- Redundant power supplies (N+1 Redundancy).
- Hot-swappable Power Supplies.
- Capable of assigning up to 4094 VLAN Tags (Trunking).
- Supports minimum Jumbo Ethernet frame size of 9000 bytes.
- Must be able to change, remove or disable the default VLAN from trunks.
- Must support at a minimum eight (8) queues for traffic prioritization.
- Must be able to assign traffic to queues.
- By VLAN tags, types of traffic or DIFFSERV Code Point.
- Set QoS by DIFFSERV Code Point based on type of traffic (e.g.: VoIP).
- Redundant Fans (N+1 Redundancy).
- Hot-swappable Fans.
- Must have a backplane that is non-blocking.
- Line-rate throughput on all ports at full capacity.
- Must be same OEM vendor as EAS devices for link layer integrity.
- Capable of filtering spanning-tree BPDU.
- Support IEEE 802.3ba 40/100GB Ethernet (Backbone/Uplink Ports) for future-proofing.

7.9.2 Uplink Port Requirements:

- A minimum of 2 each 10GBASE-X SFP+.
- Must be MACSec capable.

7.9.3 Downlink Port Requirements (10GbE to EASs)

- Must support a minimum of 24 port 10GBASE-X SFP+ (dual-speed 1/10GbE).
- Must be MACSec capable.

7.10 ACS Switch Requirements

ACS will be used by the USG to form the core backbone and Layer 2 domains of the ICAN. There will be two (2) types of ACS, a high-density and a low-density one; based on the maximum of ports that each type can support.

All ACS must support the following requirements without the addition of hardware, software, firmware or licensing:

7.10.1 General Requirements:

- Must be a modular chassis type device.
- All modules/components must be interchangeable between high-density ACS and low-density ACS chassis.
- Must be able to change, remove or disable the default VLAN from trunks.
- Capable of assigning up to 4094 VLAN Tags (Trunking).
- Supports minimum Jumbo Ethernet frame size of 9000 bytes.
- Control and data forwarding plane modules must be physically separate.
- Must support at a minimum eight (8) queues for traffic prioritization.
- Must be able to assign traffic to queues.
- By VLAN tags, types of traffic or DIFFSERV Code Point.
- Set QoS by DIFFSERV Code Point based on type of traffic (e.g.: VoIP).
- Must be flexible in compatibility.
- Module available to provide 100GB uplink/crosslink for future environments (IEEE 802.3ba).
- Module available to provide 48 port 1GbE-SFP downlinks for legacy environments.
- Internal components, other than I/O modules, must be field replaceable non-disruptively, and must support OIR.
- Must have redundant switching engines/modules.
- Capable of supporting input voltages of both 120/220 VAC or 48VDC.
- Must have N+1 redundant and hot-swappable power supplies.
- Must have N+1 redundant and hot-swappable fans.
- Capable of filtering spanning-tree BPDU.
- Capable of supporting MACSec encryption as per IEEE 802.1AE Security standard.

7.10.2 Uplink Port Requirements:

- A minimum of 2 each 100GbE ports.
- Crosslink Port (for interconnecting ACS switches):
  - A minimum of 2 each 100GbE ports for inter-connecting ACS switches.

7.10.3 Downlink Port (10GbE ports to EAS or ADS):

- High-Density ACS chassis must support a minimum of 236 10GbE SFP+ interface ports – in non-blocking configuration (Line Rate and Full Layer 2 Functionality).
- Low-Density ACS chassis must support a minimum of 80 10GbE SFP+ interface ports – in non-blocking configuration (Line Rate and Full Layer 2 Functionality).

7.11 L2BS Switch Requirements

L2BS will be used by the USG to form controlled interconnects between multiple Layer 2 Domains (Areas), usually there is only one of these devices per B/P/C/S.

All L2BS devices must support the following requirements without the addition of hardware, software, firmware or licensing:

7.11.1 General Requirements

- Must be ASLAN-Certified as Layer 2 Distribution Switch.
- Input voltage capable of both 120/220 VAC or 48 VDC.
- Redundant power supplies (N+1 Redundancy).
- Hot-swappable Power Supplies.
- Capable of assigning up to 4094 VLAN Tags (Trunking).
- Supports minimum Jumbo Ethernet frame size of 9000 bytes.
- Must be able to remove or disable the default VLAN from trunks.
- Must support at a minimum eight (8) queues for traffic prioritization.
- Must be able to assign traffic to queues.
- By VLAN tags, types of traffic or DIFFSERV Code Point.
- Set QoS by DIFFSERV Code Point based on type of traffic (e.g. VoIP).
- Redundant Fans (N+1 Redundancy).
- Hot-swappable Fans.
- Capable of filtering spanning-tree BPDU.

7.11.2 Uplink Port Requirements:

- A minimum of 2 each 10GBASE-X SFP+.
- Must be MACSec capable.

7.11.3 Downlink Port Requirements (10GbE to EASs)

- Must support a minimum of 24 port 10GBASE-X SFP+ (dual-speed 1/10GbE).
- Must be MACSec capable.

7.11.4 10GBASE-LR, SFP+ Optics (LC)

System must support following SFP+ type optics.

- 10GBASE-LR, SFP+ optic (LC), rated for 10km over Single Mode Fiber.
- 10GBASE-ER, SFP+ optic (LC), rated for 40km over Single Mode Fiber.
- 10GBASE-ZR, SFP+ optic (LC), rated for 80km over Single Mode Fiber.
- 1GBASE-SX, SFP optic (LC), for multimode fiber transmissions.
- 1GBASE-LX, SFP optic (LC), for single mode fiber transmissions.